Looking Away from the Market

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September 5th, 2026

Abstract

The Capital Asset Pricing Model (CAPM) is rejected empirically because market beta is weakly related to average returns. Using idiosyncratic news from the Dow Jones Newswire and institutional attention measures from Bloomberg search activity, I isolate states in which market-wide attention is focused on publicly available idiosyncratic news. In these states, the CAPM prices risk. Market betas explain the cross section of stock returns, and the security market line is steep and positive. In contrast, when attention is elevated but not well explained by idiosyncratic news, the beta–return relation flattens. To rationalize these findings, I develop a noisy rational expectations model in which investors choose between concentrating limited attention on noisy idiosyncratic signals or dispersing it towards broader market level learning. Because attention to idiosyncratic news crowds out private learning about systematic conditions, greater attention to idiosyncratic information increases posterior factor uncertainty. Investors require compensation for bearing this uncertainty, generating a higher equilibrium factor risk premium and a steeper security market line. Empirically, unconditional betas price returns on the more than half of trading days when attention is anchored to idiosyncratic news; the betting-against-beta premium is earned almost entirely on the remaining days.

JEL Classification: G12, G14, G11, D83

Keywords: CAPM, security market line, investor attention, idiosyncratic news, beta

*I thank Miguel Palacios, Alexander David, Charles Martineau, and many others for helpful comments. Seminar participants at the Haskayne school of Business also provided valuable feedback.
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Data for replication available at Adamupenieks.com.

1 Introduction

A persistent tension separates the theory of asset pricing from its empirical application. The capital asset pricing model (CAPM) of Sharpe (1964), Lintner (1965), and Mossin (1966) predicts that an asset's risk premium is increasing in its market beta. Despite the model's central role in academic research and practitioner discount-rate calculations (Berk and van Binsbergen, 2017), this prediction performs poorly empirically in cross-sectional tests. Recent literature offers conditional resolutions to this puzzle[1]. A common theme in these resolutions is that the beta–return relation depends on investors processing information, and several recent studies suggest that investor attention could play an important role in determining asset prices. This paper argues that the CAPM succeeds not only when investors pay more attention, but when attention is concentrated on a particular kind of information: public idiosyncratic news.

Models of attention allocation imply that investors reallocate a limited attention budget between firm-specific and systematic information (Peng and Xiong, 2006; Kacperczyk, Van Nieuwerburgh, and Veldkamp, 2016), a prediction that has recently received empirical support (Cookson, García, and Jarnecic, 2026). The asset pricing consequences of this reallocation are less explored. I address this by identifying the channel through which it operates: attention spent processing news about individual firms is time not spent learning about the state of the broader economy. As a result, when investors concentrate on idiosyncratic information they hold more uncertain beliefs about macroeconomic conditions and demand greater compensation for exposure to them. I derive closed-form conditions characterizing when this substitution increases the market price of risk and show that a single mechanism generates three predictions that I confirm empirically.

First, the security market line is state dependent: beta is priced on days when institutional attention is concentrated on firm-specific news and flat otherwise. Second, the effect is specific to market beta. Because firm-specific news carries no information about macroeconomic conditions, concentrated attention raises uncertainty about market exposure alone, and the size, value, profitability, and investment premia are unchanged across states. Third, the premium for idiosyncratic processing rises with macroeconomic uncertainty. When prior macroeconomic uncertainty is greater, so is the cost of leaving it unresolved, and high-beta assets command a larger premium when attention is concentrated on idiosyncratic news.

The theoretical model is a multi-firm variant of Grossman and Stiglitz (1980) in which in-

[1] See Tinic and West (1984), Savor and Wilson (2014), Savor and Wilson (2016), Hong and Sraer (2016), Jylhä (2018), Hendershott, Livdan, and Rösch (2020), Chan and Marsh (2022), Andrei, Friedman, and Ozel (2023), and Hasler and Martineau (2023, 2024), among others.

vestors allocate a fixed attention budget between two sources of information: public news about individual firms and private analysis of a common macroeconomic factor. The key insight is that these two activities are substitutes with asymmetric payoffs. Firm-specific news, once its market-level content is filtered out, is informationally orthogonal to the factor: it resolves uncertainty about a firm's idiosyncratic payoff and nothing else. An investor who devotes attention to it therefore holds a more precise view of individual firms and, because the budget is fixed, a less precise view of the macroeconomy. Idiosyncratic uncertainty is diversifiable and carries no price, but factor uncertainty is not, so the reallocation shifts uncertainty from a component investors are not paid to bear to one they are. To clear markets, expected returns must compensate for the factor uncertainty that remains unresolved in equilibrium, and this compensation accrues in proportion to each asset's exposure to the factor. The cross-section of expected returns therefore aligns more closely with beta, and the security market line steepens.

To bring the framework to the data, I create a daily measure of how institutional attention is divided between firm-specific and broader processing. From the Dow Jones Institutional Newswire, I extract firm-level articles and screen out systematic and earnings-related content using keyword filters and a GPT-4o-mini classifier that retains only articles about an individual firm that are not tied to that day's market developments, so that the surviving news is the orthogonal firm-specific signal of the model. From Bloomberg, I obtain firm-day institutional attention scores following Ben-Rephael, Da, and Israelsen (2017). These combine into two aggregate daily series for the Russell 3000 universe: an idiosyncratic news attention index, IAI_t, the cross-sectional average of Bloomberg attention among firms with a filtered firm-specific article on day t, and an aggregate institutional attention index, AIA_t, constructed as in Chan and Marsh (2022). Neither series identifies the headline result alone. The state variable is the residual $\widehat{u}_t$ from an expanding-window regression of AIA_t on IAI_t, estimated using only data through day t so that classification is out of sample. When $\widehat{u}_t$ is low, aggregate attention is fully accounted for by attention to firm-specific news, and attention is anchored in firm-level content. When $\widehat{u}_t$ is high, attention exceeds what firm-specific news can explain, and reflects processing directed at market-wide or macroeconomic information.

Two pieces of evidence suggest that $\widehat{u}_t$ captures the composition of attention. Fisher, Martineau, and Sheng (2022) measure macroeconomic attention independently from news coverage of aggregate topics. Table 1 shows that their index rises with $\widehat{u}_t$: on days when institutional attention exceeds what firm-specific news can account for, coverage of macroeconomic themes is elevated. This suggests the two forms of attention move as substitutes, which has recently received empirical support (Cookson, García, and Jarnecic, 2026). Further,

the residual is one of several ways to measure how concentrated attention is on firm-level content, and the results do not depend on the choice. Alternative concentration measures deliver similar state-dependent pricing.

I test the model in U.S. equity data. On the 53% of trading days on which the rolling median of $\widehat{u}_t$ lies below its trailing one-year median, beta is priced: the Fama–MacBeth slope is 12.6 basis points per day ($t = 3.00$) for value-weighted portfolios and 8.7 ($t = 2.21$) for equal-weighted portfolios, rising to 23.1 and 22.6 in the most anchored quartile, and the intercept is indistinguishable from zero, so the security market line is steep and consistent with the CAPM. On the complementary days the slope is flat. A distinct implication separates this result from the high-attention effects documented by Ben-Rephael, Carlin, Da, and Israelsen (2021): the days on which beta is most strongly priced are days on which aggregate attention AIA_t is below its sample mean while firm-news attention IAI_t is above it. What matters is the share of attention captured by firm-specific news, not its level. Replicating their expected-information-consumption framework confirms that the two channels are distinct rather than redundant: the conditional pricing result persists at full strength among firms their measure does not flag as expected to consume information.

Turning to uncertainty, the model implies that the composition channel is amplified when macroeconomic uncertainty is high. Because the premium compensates for factor uncertainty that attention to firm-specific news leaves unresolved, a larger prior uncertainty means a larger residual, so the price of beta on idiosyncratic-anchored days should rise with the level of uncertainty investors face going in. The data support this. Sorting idiosyncratic-anchored days into terciles of the prior day's uncertainty, the price of beta in the top tercile is 24.2 basis points per day ($t = 2.14$) for realized variance, 31.8 ($t = 2.92$) for VIX, and 24.1 and 24.9 ($t = 2.34$ and 2.32) for the one- and twelve-month macroeconomic uncertainty indices of Jurado, Ludvigson, and Ng (2015), against estimates between -0.4 and 12.8 in the lower terciles. The premium is a composition effect and not an uncertainty effect: on unanchored days in the same top tercile the slope is indistinguishable from zero for every proxy, and a pooled interaction of beta with the idiosyncratic-anchored-state indicator within the top tercile ranges from 24.4 to 33.8 basis points (t between 1.77 and 2.33). The steep slope is also the CAPM slope. Across the four proxies the top-tercile intercept is insignificant and the slope exceeds the average market excess return in the cell only marginally (t between 1.13 and 1.89), a tilt I disclose in Section 6.2.4. Beta is priced most steeply on the days investors are least willing to bear unresolved aggregate risk, which is the compensation the model describes.

The closest antecedent is Andrei, Friedman, and Ozel (2023), who also derive a steeper

security market line from attention to firm-level information and also find the effect strongest under high uncertainty. In their model, announcements carry systematic content, so attentive investors resolve aggregate uncertainty and the line is steep on days it is resolved. Here firm-specific news is orthogonal to the factor, so processing it leaves aggregate uncertainty unresolved and the line is steep on days it is not. The designs differ accordingly: their state variables are levels, whereas this study uses the composition of attention, on days when the level of attention is below its mean. Their model predicts a steeper line on high-uncertainty days irrespective of attention. High uncertainty alone prices nothing in this setting, and finds beta pricing in both low and high uncertainty states, conditional on attention being anchored to idiosyncratic news.

The conditional pricing result is robust to the state variables the literature has used to locate conditional CAPM performance. Savor and Wilson (2014) find that beta is priced on macroeconomic announcement days, Savor and Wilson (2016) and Chan and Marsh (2022) on earnings announcement days and the days leading into them, and Tinic and West (1984) in particular calendar months. Excluding firms within a five-day earnings announcement window barely affects the beta slope, removing macroeconomic announcement days and leading earnings announcement days leaves the magnitudes and significance levels essentially unchanged, and the attention-anchored state slope on days with no scheduled announcement is 11.5 basis points ($t = 2.54$), suggesting the finding is not an announcement premium. The result is also stable across subperiods and is not local to the firms on which the residual is measured: rebuilding beta deciles from all CRSP common stocks, beta is priced at 7.6 basis points ($t = 2.91$) on low-$\widehat{u}$ days and -2.6 on other days, among small stocks as strongly as among large ones, and in every half and third of the sample.

Hong and Sraer (2016) locate a steep line in low disagreement, Hasler and Martineau (2023) in low volatility, Hasler and Martineau (2024) in a high expected market premium, and Andrei, Friedman, and Ozel (2023) in high uncertainty resolved by attention to firm announcements. Hendershott, Livdan, and Rösch (2020) find that beta is priced overnight but not during trading hours; on low-$\widehat{u}$ days beta is priced intraday as well, at 4.6 basis points against -7.4 on other days, so that the intraday line that is flat on average is the mixture of a priced and an inverted line, while the overnight premium is present in both states. The speculative-beta mechanism of Hong and Sraer (2016) requires the flat line to be concentrated among stocks that are hard to short; it is equally flat among stocks with high and low institutional ownership. The continuous specification of Table 5 conditions on each of these objects directly. The daily Fama–MacBeth price of beta loads on the standardized five-day median of the attention residual evaluated through the prior day at -9.2 basis points per standard deviation ($t = -2.9$) for value-weighted and -7.7 ($t = -2.9$) for equal-weighted

portfolios, and this gradient moves by only a few percent under controls for disagreement, the VIX, realized variance, the option-implied premium bound of Martin (2017), which is the expected-premium object of Hasler and Martineau (2024), the variance risk premium, investor sentiment, the levels of aggregate and firm-news attention that separate this channel from Andrei, Friedman, and Ozel (2023), the supply of firm-specific news, lagged market returns, the lagged price of beta, and announcement and day-of-week dummies, and with every control entered jointly it is -10.3 ($t = -2.1$). Finally, reconstructing the attention-composition measure using only news from individual beta or size-sorted deciles shows that the conditional CAPM emerges broadly across constructions, with significant beta slopes in the majority of deciles for both sorts, indicating that the result is not driven by news from one particular segment of the cross section.

A natural concern is that the news on which attention anchors, although it survives filters for market-wide and earnings content, still conveys information about the aggregate factor. In this case, the security market line would be steep because attention resolves factor uncertainty rather than because it leaves it unresolved. Two pieces of evidence rule this out. Rebuilding the attention residual separately from the firm news that moves betas and the firm news that does not yields a price of beta of comparable magnitude in both cases, whereas a systematic-information channel would require the pricing to load on the beta-moving news alone. And although betas do rise on news days, by 0.074, the increase reflects a roughly 40 percent rise in idiosyncratic volatility against only a 5 percent rise in market covariance, so firm news makes firms noisier rather than more systematic.

The predictability is economically large. A strategy that bets on beta on low-$\widehat{u}$ days and against it otherwise earns a daily alpha of 10.6 basis points after controlling for the five **?** factors and the betting-against-beta factor, with a Sharpe ratio of 0.80, gross of transaction costs. Two features of the strategy speak to the mechanism. First, the betting-against-beta premium of Frazzini and Pedersen (2014) is earned almost entirely on the days the state classifies as unanchored to idiosyncraticl news, so a single ex-ante state separates the days on which the CAPM holds from those on which the low-beta-return anomaly appears. Second, a beta-neutral version of the strategy earns nothing beyond the unconditional betting-against-beta premium, and a market-timing regression attributes the strategy's alpha entirely to timing: the state predicts when the market premium is paid, not which stocks are mispriced, which is what a time-varying risk premium implies.

The findings also speak to a broader question about what investor attention does to asset prices. Peng and Xiong (2006) argue that limited-capacity investors exhibit category-learning behavior, choosing between processing market-wide and firm-specific information.

Veldkamp (2006b) shows that costly information acquisition generates excess comovement when investors substitute toward common signals. Kacperczyk, Van Nieuwerburgh, and Veldkamp (2016) model attention reallocation across business-cycle states and show that attention to systematic versus idiosyncratic risks varies endogenously. The mechanism developed here builds on this tradition but with a distinct emphasis: under the orthogonality structure motivated by the LLM filter, attention composition has an unconditional effect on factor uncertainty that does not rely on equilibrium feedback through prices, learning from prices, or general-equilibrium attention-supply effects.

This positioning has a deeper interpretive content. Berk and van Binsbergen (2016) document that mutual fund flows respond to CAPM alphas more strongly than to alphas from any multi-factor alternative, concluding that investors use the CAPM. Yet the unconditional CAPM is rejected in cross-sectional tests. One reconciliation is that investors form beliefs according to the CAPM but the cross-sectional pricing implication only emerges when their attention is actively engaged in processing the firm-specific information that translates beliefs into prices. The conditional result documented here is consistent with this view: the CAPM emerges in the cross section on the days when institutional attention is anchored to firm-news processing, and recedes when attention drifts elsewhere.

The remainder of the paper proceeds as follows. Section 2 sets up the model, Section 3 solves for equilibrium prices and the Fama–MacBeth slope, and Section 4 derives the comparative statics that deliver the empirical predictions. Section 5 describes the data and constructs the attention measures, Section 6 presents the empirical results, Section 7 conducts a battery of robustness tests, and Section 8 explores a simple trading strategy that exploits these periods. Section 9 concludes.

2 Model

This section presents a noisy rational expectations model of asset prices with costly attention. Investors are modeled in the tradition of Grossman and Stiglitz (1980) and Admati (1985): they are risk averse, receive Gaussian signals, and learn only imperfectly about the assets they trade. As in Veldkamp (2006b), costly information acquisition leads investors to rely more heavily on common signals, and as in Peng and Xiong (2006), investors with limited attention choose between processing market-wide and firm-specific information. The model places this choice inside a factor structure: asset payoffs load on a common factor, and each investor divides a fixed attention budget between firm-specific public news, which is orthogonal to the factor, and signals about the factor itself. When attention concentrates

on firm-specific news, the factor is learned less well and posterior factor uncertainty remains high. Because beta is compensation for bearing factor risk, investors then demand a larger premium per unit of beta, and the security market line steepens. As attention shifts toward the factor, that uncertainty resolves, the premium contracts, and the security market line flattens. I build the equilibrium in steps: the setup and information structure here, posterior beliefs, the portfolio problem, market clearing, and the equilibrium price and risk premium in Section 3, and the comparative statics that deliver the empirical predictions in Section 4.

2.1 Assets and investors

The economy lasts a single period and contains N risky assets and a risk-free asset whose return is normalized to zero. The payoff to risky asset $i \in \{1, \ldots, N\}$ is

$$d_i = \bar{d} + \beta_i f + \varepsilon_i, \tag{1}$$

where $\bar{d}$ is a known constant, $f \sim \mathcal{N}(0, \sigma_f^2)$ is the market factor, and $\varepsilon_i \sim \mathcal{N}(0, \sigma_\varepsilon^2)$ are firm-specific shocks, independent across i. The factor loading is itself uncertain,

$$\beta_i = b_i + \nu_i, \qquad \nu_i \sim \mathcal{N}(0, \sigma_\nu^2), \tag{2}$$

with b_i a known prior mean and ν_i independent across i and of (f, ε_i). I write precisions as $\tau_f \equiv \sigma_f^{-2}$, $\tau_\nu \equiv \sigma_\nu^{-2}$, and $\tau_\varepsilon \equiv \sigma_\varepsilon^{-2}$.

A continuum of investors of unit mass, indexed by $j \in [0, 1]$, has CARA preferences over terminal wealth with common risk aversion $\rho > 0$. Per-capita supply of asset i is $\bar{x} + u_i$, where $u_i \sim \mathcal{N}(0, \sigma_u^2)$ is a supply shock that prevents prices from being fully revealing.

2.2 Information

Investors receive two kinds of signals: firm-specific public news and private information about the factor.

Assumption 1 (Firm-specific public signal)**.** *For each asset i, all investors observe*

$$y_i = \varepsilon_i + \eta_i, \qquad \eta_i \sim \mathcal{N}(0, 1/\tau_y), \tag{3}$$

where η_i is independent across i and orthogonal to $(f, \beta_i, \varepsilon_i)$.

Assumption 1 treats firm-level news as informationally orthogonal to the factor. Empirically it corresponds to news classified as purely firm-specific, such as earnings releases, management changes, and product announcements, after market-level content has been filtered out. Public signals inform investors about ε_i but carry no information about f or about the loading β_i. This orthogonality is the key assumption of the model: it is what makes attention to firm news attention taken away from the factor, rather than an alternative way of learning about it, and it distinguishes the setup from standard noisy rational expectations models in which a single public signal conveys both firm-specific and factor information.

Assumption 2 (Private factor signal)**.** *Each investor j privately observes*

$$z_j = f + \xi_j, \qquad \xi_j \sim \mathcal{N}(0, 1/\tau_\xi), \tag{4}$$

with ξ_j independent across investors and orthogonal to $(f, \varepsilon_i, \nu_i, \eta_i)$.

2.3 Attention allocation

Each investor has a unit attention budget and allocates a share $\omega_j \in [0, 1]$ to firm-specific public news and the remainder $1-\omega_j$ to private factor information. Precision in each channel is linear in the attention devoted to it:

$$\tau_y = \omega_j S, \qquad \tau_\xi = (1 - \omega_j)K. \tag{5}$$

Here S is the day's supply of firm news, the precision that would be available if all attention went to firm-specific news; it can be read as the number of articles times the informativeness of each, and only the product $S\sigma_\varepsilon^2$ matters for what follows. K is private research capacity, the precision available if all attention went to the factor, and only $K\sigma_f^2$ matters. Higher ω_j means attention concentrated on firm-specific news; lower ω_j means attention diverted to the factor. Attention is chosen before any signal is observed, and I focus on the symmetric equilibrium in which all investors choose the same ω.

2.4 Posterior beliefs

All primitives and signals are jointly Gaussian, so posteriors follow from normal updating.

Lemma 1 (Posteriors)**.** *Under Assumptions 1 and 2, the posterior distributions are mutually*

independent and given by

$$f \mid z_j \sim \mathcal{N}\big(\mu_{f,j},\, V_f\big), \qquad V_f = \frac{1}{\tau_f + (1-\omega)K}, \qquad \mu_{f,j} = V_f \tau_\xi z_j, \tag{6}$$

$$\beta_i \mid y_i \sim \mathcal{N}\big(b_i,\, \sigma_\nu^2\big), \tag{7}$$

$$\varepsilon_i \mid y_i \sim \mathcal{N}\big(\mu_{\varepsilon,i},\, V_\varepsilon\big), \qquad V_\varepsilon = \frac{1}{\tau_\varepsilon + \omega S}, \qquad \mu_{\varepsilon,i} = V_\varepsilon \tau_y y_i. \tag{8}$$

The proof is in Appendix A. In each case posterior precision is prior precision plus signal precision, and the posterior mean is a precision-weighted average of the prior mean, which is zero, and the signal. Three features carry forward. First, the posterior of f is determined entirely by the private signal: under Assumption 1 the public signals are independent of (f, z_j), so they drop out of the factor posterior. Second, the posterior of β_i equals its prior, because no signal depends on ν_i; no amount of attention reduces loading uncertainty. Third, the posterior variances V_f and V_ε depend on ω but not on the signal realizations, which is why they are common across investors and why an investor can evaluate them when choosing attention. The posterior means differ across investors only through the private noise ξ_j, and by the law of total variance $\mathrm{Var}(\mu_{f,j}) = \sigma_f^2 - V_f$, a fact used below.

3 Equilibrium

3.1 Portfolio choice

Investor j chooses holdings q_j to maximize $\mathbb{E}_j[-\exp(-\rho W_j)]$ with $W_j = q_j'(d - p)$, where d and p are the $N \times 1$ vectors of payoffs and prices. Because the information set $\mathcal{I}_j = \{z_j, y_1, \ldots, y_N\}$ renders d conditionally Gaussian, with mean $\mu_j = \mathbb{E}[d \mid \mathcal{I}_j]$ and covariance $\Sigma = \mathrm{Var}(d \mid \mathcal{I}_j)$, terminal wealth is Gaussian and the objective reduces to the mean-variance criterion $q_j'(\mu_j - p) - \frac{\rho}{2} q_j' \Sigma q_j$. The first-order condition equates the expected profit per share to the marginal risk cost of one more share, and demand is

$$q_j = \frac{1}{\rho} \Sigma^{-1}(\mu_j - p). \tag{9}$$

By Lemma 1, and because ν_i is independent of f with mean zero so that $\mathbb{E}[\beta_i f \mid \mathcal{I}_j] = b_i \mu_{f,j}$, investor j's forecast of asset i is

$$\mu_{j,i} = \bar{d} + b_i \mu_{f,j} + \mu_{\varepsilon,i}. \tag{10}$$

3.2 Posterior covariance

The posterior covariance describes the risk that remains after learning and which of it is shared across assets. Writing the payoff as $d_i = \bar{d} + b_i f + \nu_i f + \varepsilon_i$ and subtracting the forecast in (10), the forecast error decomposes into three surprises,

$$d_i - \mu_{j,i} = \underbrace{b_i(f - \mu_{f,j})}_{\text{factor}} + \underbrace{\nu_i f}_{\text{loading}} + \underbrace{\varepsilon_i - \mu_{\varepsilon,i}}_{\text{idiosyncratic}}. \tag{11}$$

The factor surprise is common to all assets and scaled by each asset's loading; the loading and idiosyncratic surprises are specific to asset i. Multiplying out the cross-products and taking conditional expectations, all cross terms vanish because ν_i and ε_i have mean zero and the three components are independent (Appendix B), leaving

$$\Sigma_{ik} = b_i b_k V_f + \mathbf{1}\{i = k\}\Big(V_\varepsilon + \sigma_\nu^2\,\mathbb{E}[f^2 \mid \mathcal{I}_j]\Big), \qquad \mathbb{E}[f^2 \mid \mathcal{I}_j] = V_f + \mu_{f,j}^2. \tag{12}$$

Two assets covary only because both load on the factor, and the covariance is the product of their loadings times the posterior factor variance. The diagonal adds the variance the investor cannot learn away: the unlearned idiosyncratic variance and the variance created by not knowing the loading.

The diagonal term contains $\mu_{f,j}^2$ and so depends on investor j's own signal draw. To obtain a single covariance matrix common to all investors, which is required for the aggregation below, I replace $\mu_{f,j}^2$ by its ex-ante expectation. Since $\mathrm{Var}(\mu_{f,j}) = \sigma_f^2 - V_f$, the expectation of $V_f + \mu_{f,j}^2$ is σ_f^2, and the V_f terms cancel. Define the residual non-factor variance

$$V_\varepsilon^\star \equiv V_\varepsilon + \sigma_\nu^2 \sigma_f^2, \tag{13}$$

so that

$$\Sigma = V_f\, bb' + V_\varepsilon^\star\, I. \tag{14}$$

The covariance has a rank-one factor component and a diagonal residual component. The averaging affects only the common diagonal term and does not touch the cross-sectional structure that determines the price of beta.

3.3 Market clearing

Aggregate demand integrates individual demands over the continuum. Because every investor chooses the same ω in the symmetric equilibrium, Σ and p are common, and only the

forecasts μ_j differ:

$$\int_0^1 q_j\, dj = \frac{1}{\rho}\,\Sigma^{-1}\big(\bar{\mu} - p\big), \qquad \bar{\mu} \equiv \int_0^1 \mu_j\, dj. \tag{15}$$

Integrating (10) over investors, the constant and the public-news term are the same for everyone, and the private factor forecasts average out their noise: $\int_0^1 \mu_{f,j}\, dj = V_f \tau_\xi \int_0^1 (f + \xi_j)\, dj = V_f \tau_\xi f \equiv \bar{\mu}_f$. Hence $\bar{\mu}_i = \bar{d} + b_i \bar{\mu}_f + \mu_{\varepsilon,i}$. Setting aggregate demand equal to the supply vector $\bar{x}\mathbf{1} + u$ and solving,

$$p = \bar{\mu} - \rho\,\Sigma\,(\bar{x}\mathbf{1} + u). \tag{16}$$

The ith entry of $\Sigma(\bar{x}\mathbf{1} + u)$ follows from (14): using $\sum_k b_k = N\bar{b}$ with $\bar{b} \equiv N^{-1}\sum_i b_i$, the supply term is $V_f b_i (N\bar{b}\bar{x}) + V_\varepsilon^\star \bar{x}$ and the noise term is $V_f b_i (b'u) + V_\varepsilon^\star u_i$ (Appendix C).

3.4 Equilibrium price

Lemma 2 (Prices)**.** *The equilibrium price of asset i is*

$$p_i = \bar{d} + b_i \bar{\mu}_f + V_\varepsilon \tau_y\, y_i - \rho V_f\, b_i\, (N\bar{b}\,\bar{x}) - \rho V_\varepsilon^\star\, \bar{x} - \rho V_f\, b_i\, (b'u) - \rho V_\varepsilon^\star\, u_i, \tag{17}$$

where $\bar{\mu}_f = V_f \tau_\xi f$ is the average posterior mean of the factor across investors.

Each term has a direct reading. The first is the baseline payoff. The second is the market's average assessment of the factor, scaled by the asset's exposure to it. The third is the price response to the asset's own public news; the coefficient $V_\varepsilon \tau_y = \omega S/(\tau_\varepsilon + \omega S)$ is the fraction of the news that is impounded into the price and rises with the attention devoted to firm news. The fourth is the factor risk discount: aggregate holdings $\bar{x}$ per asset carry total factor exposure $N\bar{b}\bar{x}$, the risk of that position scales with posterior factor uncertainty V_f, and asset i bears a share of the discount proportional to its loading b_i. A lower price here is a higher expected return. The fifth is the residual risk discount, common to all assets because $V_\varepsilon^\star$ is. The last two are the price effects of supply noise, one transmitted through the factor and one asset-specific, which prevent prices from revealing the average factor signal.

3.5 Risk premium and the security market line

Taking unconditional expectations of (17), the factor forecast, the news, and the supply shocks all have mean zero, so

$$\mathbb{E}[d_i - p_i] = \rho V_f\, b_i\, (N\bar{b}\,\bar{x}) + \rho V_\varepsilon^\star\, \bar{x}. \tag{18}$$

Both discounts in the price become positive premia in the return. The expected excess return is linear in the prior loading b_i with slope $\rho V_f N\bar{b}\bar{x}$ and intercept $\rho V_\varepsilon^\star \bar{x}$: this is the model's security market line. The slope is the price of factor risk, and it is proportional to the posterior variance of the factor, the object that attention allocation moves. The intercept is compensation for residual risk, which is common across assets and therefore invisible to a cross-sectional regression on betas.

3.6 The Fama–MacBeth slope

The econometrician does not observe b_i. Loadings are estimated by a T-period rolling regression of realized returns on the factor, $\hat{\beta}_i = \beta_i + e_i = b_i + \nu_i + e_i$, where e_i is sampling error with variance V_e, and the price of beta is the cross-sectional slope of expected returns on $\hat{\beta}_i$. Treating b_i as a cross-sectional draw with variance σ_b^2, independent of ν_i and e_i, the covariance of expected returns with $\hat{\beta}_i$ is $\lambda^A \sigma_b^2$ with $\lambda^A \equiv \rho V_f N\bar{b}\bar{x}$, while the variance of $\hat{\beta}_i$ is $\sigma_b^2+\sigma_\nu^2+V_e$. With $s_\beta^2 \equiv \sigma_b^2+\sigma_\nu^2$ the cross-sectional variance of true betas, the Fama–MacBeth slope is

$$\lambda^{FM}(\omega) = A(\omega)\cdot \rho V_f(\omega)\, N\bar{b}\,\bar{x}, \qquad A(\omega) = \frac{s_\beta^2 - \sigma_\nu^2}{s_\beta^2 + V_e(\omega)}, \qquad V_e(\omega) = \frac{\sigma_\varepsilon^2 + 1/(\omega S)}{T\sigma_f^2}, \tag{19}$$

where $A(\omega) \in (0,1)$ is the errors-in-variables attenuation (Appendix D). Realized returns equal expected returns plus a mean-zero surprise, so the slope on any single day is λ^{FM} plus a term driven by that day's factor and supply realizations. Averaging over days removes it provided the attention state does not forecast the market and the beta estimation window ends before the return day, both of which the empirical design imposes.

4 Comparative Statics

The empirical content of the model lies in how the price of beta responds to the allocation of attention.

4.1 The risk-premium channel

Proposition 1 (Risk-premium channel)**.** *Posterior factor uncertainty is strictly increasing in attention to firm-specific news:*

$$\frac{dV_f}{d\omega} = KV_f^2 > 0. \tag{20}$$

Proof. From $V_f = (\tau_f + (1-\omega)K)^{-1}$, differentiating gives $dV_f/d\omega = -V_f^2 \cdot d[\tau_f + (1-\omega)K]/d\omega = KV_f^2 > 0$. □

Proposition 1 holds unconditionally: it does not depend on how ω is chosen, on parameter values, or on any equilibrium condition. Under Assumption 1, firm-specific news carries no factor information, so the only channel through which investors learn about f is the private signal. Each unit of attention allocated to firm news is one unit lost from factor learning, and the posterior variance of the factor rises in ω at the rate KV_f^2.

4.2 Conditional steepness of the security market line

Proposition 2 (Conditional CAPM steepness)**.** *The Fama–MacBeth slope is strictly increasing in the share of attention allocated to firm-specific news:*

$$\frac{d\lambda^{FM}}{d\omega} = \rho N\bar{b}\,\bar{x}\left[A(\omega)\,KV_f^2 + V_f\,\frac{dA}{d\omega}\right] > 0, \qquad \frac{dA}{d\omega} = -\frac{\sigma_b^2}{\left(s_\beta^2 + V_e\right)^2}\,\frac{dV_e}{d\omega} > 0. \tag{21}$$

The first term in the bracket is the risk-premium channel of Proposition 1. Every investor holds $\bar{x}$ shares of each asset on average, a position whose factor exposure is $N\bar{b}\bar{x}$ and whose risk is $(N\bar{b}\bar{x})^2 V_f$; asset i's marginal contribution to that risk is $b_i(N\bar{b}\bar{x})V_f$, so it earns $\rho V_f b_i N\bar{b}\bar{x}$ in excess of the riskless rate, and this compensation grows with V_f. The second term is a measurement channel: greater attention to firm news raises τ_y, lowers the residual variance of returns, and hence lowers the sampling variance V_e of estimated betas, which reduces attenuation. Both channels are positive, so the slope is increasing in ω. In the empirical implementation the beta estimation window predates the attention state, so V_e is effectively fixed across states and the measured effect is the risk-premium channel.

4.3 Amplification by prior uncertainty

The strength of the composition channel depends on how much factor uncertainty there is to leave unresolved.

Corollary 1 (Amplification by prior factor uncertainty)**.** *Both the price of factor risk and its sensitivity to attention are increasing in prior factor variance:*

$$\frac{\partial V_f}{\partial \sigma_f^2} = \left(\frac{V_f}{\sigma_f^2}\right)^2 > 0, \qquad \frac{\partial}{\partial \sigma_f^2}\left(\frac{dV_f}{d\omega}\right) = 2KV_f \frac{\partial V_f}{\partial \sigma_f^2} > 0. \tag{22}$$

Consequently the anchored-day price of beta $\lambda^A = \rho V_f N \bar{b} \bar{x}$ and the attention gradient $d\lambda^A/d\omega$ are both larger when investors enter the day with greater uncertainty about the market factor.

Proof. Write $V_f = (\sigma_f^{-2} + (1-\omega)K)^{-1}$. Differentiating with respect to σ_f^2 gives $\partial V_f/\partial \sigma_f^2 = V_f^2 \sigma_f^{-4} = (V_f/\sigma_f^2)^2 > 0$. Since $dV_f/d\omega = KV_f^2$ by Proposition 1, its derivative with respect to σ_f^2 is $2KV_f \cdot \partial V_f/\partial \sigma_f^2 > 0$. The statements about λ^A and $d\lambda^A/d\omega$ follow because both are proportional to V_f and KV_f^2 respectively, with positive constants. □

The premium on anchored days compensates for the factor uncertainty that attention to firm-specific news leaves unresolved, and a larger prior variance means a larger residual: the same allocation of attention leaves more uncertainty standing when there is more to begin with. Empirically, prior factor uncertainty is observable through the volatility and uncertainty indices measured before the return day, so Corollary 1 predicts that the anchored-day price of beta rises with the prior day's level of market uncertainty, a prediction Section 6.2 tests directly. The corollary also separates the model from accounts in which uncertainty flattens the line: here uncertainty is the quantity being priced, so more of it steepens the anchored-day line rather than eroding it.

When investors concentrate attention on firm-specific news, they bear more factor uncertainty in equilibrium, and expected returns must compensate for it: cross-sectional pricing aligns more sharply with betas and the security market line steepens. When attention shifts toward the factor, posterior factor uncertainty falls, the factor premium contracts, and the line flattens. Across days, variation in ω can be driven by objects the model takes as given, such as the supply of firm news S, private research capacity K, or the composition of investors. Appendix E endogenizes the allocation and shows that the investor's optimum is interior, so that attention is always divided between the two channels; the propositions above hold for any ω and do not depend on that choice. The model does not pin down the daily determinants of ω, but it is clear about the implication once ω is observed: days on which

attention is anchored on firm-specific news produce steep security market lines, and days on which it is not produce flat ones.

5 Data and Variable Construction

5.0.1 News data

As opposed to Savor and Wilson (2014), Chan and Marsh (2022) and Andrei, Friedman, and Ozel (2023) who focus their analysis on days with scheduled market-relevant announcements, I focus my analysis on news that is firm-specific and not regularly scheduled. My news comes from the Dow Jones Institutional Newswire (DJIN) accessed through ProQuest TDM Studio. To search for news, I create a string containing the name of every company listed in the Russell 3000 database since 2009 when DJIN coverage starts. I then perform a search through the DJIN for articles whose title contains at least one unique CRSP company name. I keep only articles for which a company name is included in the title of the newswire article, as Aït-Sahalia, Li, and Li (2024) find these to be most relevant to the firm, and likely to cause immediate jumps in stock prices. To avoid capturing news about earnings, since they are known to also convey systematic information, I exclude articles for which the title contains the word stem 'earning', as well as other words commonly associated with articles written about earnings reports. I further exclude news whose headlines include the words "Buzz", "Wrap up", and any other commonly repeated keywords associated with summaries of previously posted actual news. A full list of these filters is provided in the Appendix.

Newswire coverage is often noisy, and only weakly informative about firm fundamentals, making it difficult to locate relevant articles. A further challenge is to separate truly idiosyncratic news items from broad or market-wide news. Recent advances in large language models permit more refined filtering of such news, allowing me to isolate truly idiosyncratic news with greater precision. To refine the set of news articles used in the analysis, and focus on articles that are directly related to an individual firm, I use GPT-4o mini provided by TDM Studio to filter out news which is systematic or not relevant to individual firms. I use a prompt that instructs the model to use only information included in the text, and opt for a straightforward prompt to establish a foundation for the results. Nonetheless, employing more detailed prompts could allow for the extraction of more tailored information. The purpose of this step is to keep only articles written about individual firms and singular (as opposed to systematic) events. I ask GPT-4o mini to read each article and classify whether the article is related to one firm or many, and whether the article is written about one event

or many (market-wide) events. The exact prompt is shown in the Appendix. Once ChatGPT has read and labelled each of the initial 902,873 articles, I keep only those labelled as 'singular' and 'idiosyncratic'. The final GPT-labelled news corpus consists of 512,733 news articles.

5.0.2 Attention data

I follow Ben-Rephael, Carlin, Da, and Israelsen (2021) in constructing daily stock-level attention measures from Bloomberg terminals. Bloomberg records the number of times terminal users search for or read news articles on particular stocks. Bloomberg aggregates these numbers into hourly counts and creates an attention score by comparing the average hourly count during the previous eight hours to all hourly counts over the previous month for that same stock. They assign a score of 0, 1, 2, 3, or 4 if the rolling average is less than 80%, between 80% and 90%, between 90% and 94%, between 94% and 96%, and greater than 96% of the hourly counts over the previous 30 days. They then aggregate these scores up to a daily frequency by taking a maximum of all hourly scores through the day.

5.0.3 Stock return data

I retrieve excess market returns from Kenneth French's website. Following (Savor and Wilson, 2014) among others, I construct ten beta monthly-sorted portfolios using U.S. common stocks from CRSP which have a share code of 10 or 11, that match the sample of stocks with available attention data. The number of unique stocks is 2469. Daily stock betas are obtained from a rolling regression of the past 252 daily excess stock returns on the excess market returns. At the beginning of each month, stocks are sorted into one of ten beta-deciles, and the daily value-weighted and equal-weighted excess return of each decile portfolio is computed over the month.

For each portfolio, I estimate daily rolling betas by regressing the past 252 excess returns of each portfolio on the market return. These ten beta-sorted portfolios form my main test assets. In robustness tests, I also consider individual stocks and characteristic-sorted portfolios.

5.0.4 Defining high attention concentration periods

To identify periods where attention is concentrated on firm-specific news, and bring the model's attention-composition parameter ω to the data, I require a daily measure of how

heavily institutional attention is tilted toward firm-specific news. I first develop a news attention index (IAI). Using Bloomberg query scores each day, I assign the attention value from 0-4 to individual stocks which had firm-specific news arrive through the Dow Jones Institutional Newswire, and take the daily average across only firms which have a news items on each day. As an additional step, I also recreate the aggregate institutional investor attention measure developed in Ben-Rephael, Da, and Israelsen (2017) using all Russell 3000 firms and create an average daily *AIA* following Chan and Marsh (2022), taking the mean level of attention paid to all firms for day t. 2469 unique stocks for which attention data and idiosyncratic news articles are available are used to create these attention measures. The sample runs from 2010 to 2025. I run the following daily expanding regression:

$$\mathrm{AIA}_t = \alpha_{t-1} + \beta_{t-1}\,\mathrm{IAI}_t + u_t, \tag{23}$$

I estimate $(\widehat{\alpha}_{t-1}, \widehat{\beta}_{t-1})$ using only information available up to day $t-1$ in an expanding window, and define $\widehat{u}_t$ as the resulting out-of-sample residual. $\widehat{u}_t$ is the component of total institutional attention that is not explained by contemporaneous idiosyncratic-attention intensity, or 'unanchored attention': marketwide attention not accounted for by firm-level news attention. Figure 2 plots the time series of the residual. Its lowest readings, when attention is most anchored on firm-specific news, occur in the calm years of 2016 and 2017 and during the firm-news-intensive period of mid-2021; its highest readings occur during the 2022 tightening cycle, when macroeconomic news dominated, and around the volatility episode of August 2024.

The expanding-window construction serves two purposes. First, because each day's residual uses only information available through the prior day, the measure is free of look-ahead bias: any relationship documented between the residual and the conditional slope of the security market line cannot be an artifact of using future data to construct the regressor. Second, the out-of-sample residual measures the innovation in non-news attention: how much higher or lower aggregate attention is than the historical news–attention relationship would predict, rather than its absolute level. This is the appropriate object for the test, which concerns cross-day variation in the slope of the security market line rather than its level. The model predicts that the line is steeper on days when attention is composed toward firm-specific news (low residual) than on days when it is tilted toward market-wide learning (high residual). Provided the estimated news–attention coefficient is stable over the sample, the expanding residual coincides with a contemporaneous orthogonalization of attention, and its time variation reflects shifts in attention composition rather than drift in the estimated relationship. Section 7.7 considers alternative measures of attention concentration on idiosyncratic news.

Days when this residual is low are days when average institutional firm-level attention is well explained by firm-specific news. Figure 1 plots the macroeconomic attention index across quartiles of the residual: it is lowest on the days most anchored on firm-specific news and rises as attention detaches from firms, by about 0.3 standard deviations from the most to the least anchored quartile, so the residual is related to macroeconomic attention in the direction the mechanism implies but is not identified by it. Panel B of Table 1 shows how other state variables behave within quartiles of the residual measure. Macroeconomic attention of Fisher, Martineau, and Sheng (2022) rises monotonically from the lowest residual quartile to the highest, suggesting that investors substitute idiosyncratic learning for macroeconomic learning, in line with the empirical findings of Cookson, García, and Jarnecic (2026). Building from the theory section, these should be days where investors demand a risk premium for market factor exposure. The empirical results test the CAPM on days when this residual is low.

6 Empirical Analysis

6.1 Testing the CAPM across attention states

The theoretical model predicts that the beta-return relationship should be increasing in the level of attention devoted to processing firm-specific news. Figures 3 and 4 depict the main result, plotting the security market line for both value-weighted and equal-weighted portfolios separately for quantiles of $\widehat{u}_t$. For the figures, I use unconditional full-sample betas with excess returns averaged over the respective sample periods, following Savor and Wilson (2014) and Hendershott, Livdan, and Rösch (2020). When $\widehat{u}_t$ falls below its trailing one-year 50th percentile, aggregate institutional attention is well explained by idiosyncratic news attention, and the security market line slopes sharply upward. On the complementary set of days, the SML is flat.

I push the analysis further using both pooled panel regressions and Fama–MacBeth cross-sectional regressions. For the pooled specification, following Savor and Wilson (2014), Chan and Marsh (2022), and Hendershott, Livdan, and Rösch (2020), I estimate

$$r_{p,t+1} = \alpha + b_1 \widehat{\beta}_{p,t} + b_2 \mathbb{1}\{\text{State}_{t+1}\} + b_3 \widehat{\beta}_{p,t} \mathbb{1}\{\text{State}_{t+1}\} + \varepsilon_{p,t+1}, \tag{24}$$

where $r_{p,t+1}$ is the excess return of portfolio p on day $t+1$, $\widehat{\beta}_{p,t}$ is the rolling one-year portfolio beta estimated from the past 252 daily excess returns of each portfolio on the excess market return, and $\mathbb{1}\{\text{State}_{t+1}\}$ is a dummy variable equal to one when the five-day

rolling median of $\widehat{u}$, evaluated through day t, falls below (or above) the indicated trailing one-year quantile cutoff of the daily residual, also evaluated through day t. The classification of day $t+1$ therefore uses only information available at the end of day t, mirroring the timing of the rolling beta, so that the state is known before the return is earned. The five-day median and the one-day lag each serve a distinct purpose. The daily residual measures a persistent attention regime with classification noise around it, and the median filters out marginal days admitted by that noise; the lag ensures that the conditional pricing results are predictive rather than a reflection of same-day co-movement between attention and returns. The Internet Appendix reports the full grid of smoothing windows and signal lags and shows that the results below are insensitive to these choices within the horizon over which the attention regime persists. Because the five-day median is less variable than the daily residual, it falls below the trailing 25th percentile of the daily distribution on fewer than a quarter of days, so the bottom-quartile state comprises 18.0% of the sample rather than 25%. The coefficient b_1 captures the unconditional beta–return relation on non-state days, b_2 captures the state-minus-other-days alpha, and b_3 measures the incremental change in the slope of the security market line on state days relative to all other days. Standard errors are clustered at the daily level.

For the Fama–MacBeth specification, I estimate daily cross-sectional regressions of portfolio excess returns on rolling portfolio betas separately within each attention state:

$$r^{S}_{p,t+1} = a^{S} + b^{S}\,\widehat{\beta}_{p,t} + \varepsilon^{S}_{p,t+1}, \tag{25}$$

where the superscript S indexes the attention state, and $\widehat{\beta}_{p,t}$ is the prior-day rolling portfolio beta. I report time-series averages of the daily slope coefficients b^S with Newey–West t-statistics computed with five lags on the time series of the estimated daily coefficients. I report results for four attention states defined by the rolling quantile cutoffs of $\widehat{u}$: the bottom quartile (0–25%), the bottom half (<50%), the top half (>50%), and the top quartile (>75%).

Table 2 presents the results in the layout of Savor and Wilson (2014): for each set of test assets, the Fama–MacBeth estimates on low-$\widehat{u}$ days and on other days, their difference, and the pooled regression in which the low-$\widehat{u}$ dummy and its interaction with beta are estimated jointly. The test assets are formed from the firms with institutional attention data, on which $\widehat{u}$ is measured; Section 7.8 repeats the exercise on all CRSP common stocks.

For value-weighted beta-sorted portfolios (Panel A of Table 2), beta is priced on low-$\widehat{u}$ days and not otherwise. The Fama–MacBeth slope is 12.6 basis points per day ($t = 3.00$) on the 1,668 days below the trailing median of $\widehat{u}$, with an intercept of -3.0 ($t = -0.91$) that is indistinguishable from the zero the CAPM predicts, and -3.4 ($t = -0.71$) on the

remaining 1,493 days, where the intercept absorbs the average return. The difference of 16.1 basis points ($t = 2.53$) is the state contrast, and the pooled interaction reproduces it: 18.7 basis points ($t = 3.02$) against an unconditional beta coefficient of -4.4 ($t = -0.96$). The bottom-quartile row shows that the pricing strengthens with the depth of anchoring: on the 601 days below the trailing 25th percentile the slope is 23.1 basis points ($t = 2.85$), nearly twice the below-median estimate, with an intercept of -7.0 ($t = -1.07$). Equal-weighted portfolios (Panel B) give the same pattern at lower amplitude, 8.7 ($t = 2.21$) against -4.2, with a pooled interaction of 13.4 ($t = 2.31$). Individual stocks (Panel C), for which errors in variables in the betas attenuate the slopes, price beta at 6.5 basis points ($t = 2.00$) on low-$\widehat{u}$ days and -3.0 otherwise, a difference of 9.4 ($t = 1.85$), with a pooled interaction of 9.3 ($t = 1.87$).

Across the three sets of test assets the security market line is steep on low-$\widehat{u}$ days and flat on the others, the low-$\widehat{u}$ intercepts are statistically zero for the portfolio tests, and the Fama–MacBeth differences and the pooled interactions agree to within a few basis points. The CAPM prices risk when institutional attention is anchored on firm-specific news, even though the level of attention on those days is below average, and the pricing is strongest where the anchoring is deepest, which is the pattern the model's comparative statics predict. The unconditional slope, pooling all days, is about 5 basis points and insignificant, so the days on which the CAPM holds are the low-$\widehat{u}$ days and the days on which it fails are the rest, in roughly equal measure.

6.2 Uncertainty and the Attention Premium

Corollary 1 predicts that the composition premium documented above should be increasing in the level of prior factor uncertainty: on days when institutional attention is anchored to firm-specific news, beta should command a higher price the greater the unresolved uncertainty about the common factor that investors carry into the day. This section tests that implication.

The sensitivity of posterior factor uncertainty to attention composition is $dV_f/d\omega = KV_f^2$, and V_f is increasing in prior factor variance, so the same shift of attention toward firm-specific news forgoes factor learning that is more valuable, and moves the equilibrium price of beta by more, when background factor uncertainty is high. A second margin follows from the linearity of the premium in risk aversion: the composition effect on the price of beta scales with ρ, so it should also be amplified when the compensation investors demand per unit of risk is elevated. The model assigns no such role to sentiment or to belief dispersion, so

those conditioning variables serve as predicted nulls. The prediction under test is therefore a set of cross-partials rather than a level effect: the composition premium should be larger following states of high uncertainty and high risk aversion, and elevated uncertainty without the firm-anchored attention state should not price beta.

6.2.1 Design and main result

I condition the firm-anchored state on the prior day's level of uncertainty, measured four ways: trailing 21-day realized market variance (physical and backward looking), the VIX (option implied and forward looking), and the macroeconomic uncertainty index of Jurado, Ludvigson, and Ng (2015) at the one-month and twelve-month horizons (fundamental, estimated from a large panel of macroeconomic series, and available only monthly). Every conditioning variable is lagged: the attention state is classified from information through the close of day $t-1$, the daily uncertainty proxies are $t-1$ values, and the monthly indices are mapped by prior calendar month. Tercile breakpoints are computed within sample. Within the state, I estimate the Fama-MacBeth price of beta separately in each uncertainty tercile; the placebo repeats the top-tercile estimation on unanchored days, where the model implies the amplification has nothing to amplify.

Table 3, Panel A, reports the result. The price of beta on firm-anchored days rises from small and mostly insignificant values in the two lower terciles to 24.2 basis points ($t = 2.14$) in the top tercile of realized variance, 31.8 ($t = 2.92$) for the VIX, 24.1 ($t = 2.34$) for JLN at h=1, and 24.9 ($t = 2.32$) at h=12. The agreement across proxies is close in magnitude despite the proxies sharing little beyond the concept: a backward-looking return-based measure, a forward-looking option-implied measure, and a monthly index estimated from macroeconomic fundamentals place the amplified premium within eight basis points of one another. The pattern is a threshold rather than a gradient. The middle tercile is insignificant for every proxy, and condensing the cut concentrates the effect further: at quintiles, the top-bin slope reaches 52.2 basis points ($t = 2.89$) for the VIX on 267 days, while lower bins remain flatter. The premium is not increasing smoothly in uncertainty; it is concentrated on days following the most elevated uncertainty readings.

The placebo column establishes that uncertainty is an amplifier and not a channel. On unanchored days, the top uncertainty tercile prices beta at -8.6 basis points ($t = -0.82$) for realized variance, -2.8 ($t = -0.25$) for the VIX, and effectively zero for both JLN horizons. Elevated uncertainty without the firm-anchored attention state produces nothing. The result survives the removal of 2020 and 2022 in full: the top-tercile slopes are 19.9 ($t = 2.81$), 21.5 ($t = 2.92$), 16.2 ($t = 2.16$), and 16.1 ($t = 2.28$) across the four proxies, and the top-tercile

state days span more than thirty calendar quarters with no quarter contributing more than ten percent, so no single episode carries the cell. A median split of each proxy, which involves no tercile choice, delivers the same conclusion in coarser form: the high half prices at 18.9 to 21.7 basis points for the daily proxies with the low halves quiet, and the high-minus-low difference is 17.7 basis points ($t = 2.09$) for realized variance.

6.2.2 Composition conditions, uncertainty scales

A natural question is whether these cells reveal an uncertainty effect rather than an attention effect. It is possible that beta is simply priced when uncertainty is high, and the attention state is incidental. Panel B of Table 3 rejects this interpretation within the top tercile itself. Restricting the sample to top-tercile days of each uncertainty proxy, I estimate the pooled panel regression

$$r^e_{p,t} \;=\; a \;+\; b_1\,\beta_{p,t} \;+\; b_2\,D_t \;+\; b_3\,\beta_{p,t}\,D_t \;+\; \varepsilon_{p,t}, \tag{26}$$

where $r^e_{p,t}$ is the day-t excess return of beta decile p, $\beta_{p,t}$ is its lagged rolling beta, D_t is the firm-anchored state indicator classified through the close of day $t-1$, and standard errors are clustered by day. The coefficient b_1 measures the price of beta on unanchored high-uncertainty days, and b_3 measures the additional price of beta when attention is firm-anchored, holding the uncertainty regime fixed. The estimates of b_3 are 33.8 basis points ($t = 2.33$) for realized variance and 30.3 ($t = 2.07$) for the VIX, while b_1 is never significant in any column and its point estimates are negative throughout. The same holds inside every uncertainty regime, not only the top tercile: b_3 is positive in all twelve tercile-by-proxy cells, and the daily Fama-MacBeth slope loads negatively on the lagged attention residual within every tercile, including the calmest third of the sample, where the coefficient is -7.4 basis points per standard deviation ($t = -2.49$). The two conditioning variables are also orthogonal by measurement: the correlation between the attention residual and option-implied variance is 0.009, so the state is not an uncertainty proxy and its tercile cells are populated evenly.

The division of labor is therefore the one the model specifies. Composition conditions the pricing of beta at every level of uncertainty; uncertainty scales the premium that composition commands. Removing the high-uncertainty days from the sample leaves a below-median slope near its unconditional value; removing the attention measure and conditioning on uncertainty alone leaves nothing.

6.2.3 The risk-aversion margin

The second cross-partial operates through ρ. The variance risk premium, the gap between option-implied and realized variance, is the standard proxy for the effective price of risk (Bollerslev, Tauchen, and Zhou, 2009), and because the wedge is mechanically related to the uncertainty level, I hold the level fixed: within the anchored top-SVIX^2-tercile cell, days are split at the median variance risk premium, where SVIX^2 is the option-implied lower bound on the conditional equity premium of Martin (2017). Panel C reports the result. The high-wedge half prices beta at 51.8 basis points ($t = 3.60$); the low-wedge half prices it at 10.4 ($t = 0.63$). Excluding 2020 and 2022, the split is 36.0 ($t = 3.69$) against 14.7 ($t = 1.25$). Holding the quantity of uncertainty fixed, the composition premium concentrates on days when the price of risk is elevated, which is what linearity in ρ implies. The predicted nulls behave as predicted: neither investor sentiment nor belief dispersion amplifies the composition premium, with the sentiment interaction indistinguishable from zero.

The market's own premium concentrates in the same gated cells. The average market excess return on anchored top-tercile days is 16.6 basis points per day ($t = 3.59$), against 5.8 ($t = 0.82$) on unanchored top-tercile days, and 14.2 ($t = 4.26$) excluding 2020 and 2022. The time series and the cross section agree on where compensation for systematic risk is earned.

6.2.4 Discussion

Two features of Table 3 fall outside the model's predictions, and I report them plainly. First, the macroeconomic indices price the *lowest* tercile as well: 12.8 basis points ($t = 2.95$) at h=1 and 10.8 ($t = 2.51$) at h=12, a pattern that is robust across finer grids but absent from the daily market-based proxies. At monthly frequency the relation is U-shaped, with pricing at both extremes of the macroeconomic uncertainty distribution and none in the middle, whereas the daily proxies show a pure threshold. Second, in the amplified cells the security market line is steeper than the market's own premium: top-tercile slopes run roughly 1.6 to 1.9 times the average market excess return in the same cells, with correspondingly negative zero-beta intercepts, significantly so in the high-wedge cell of Panel C (-28.6, $t = -2.21$). High-beta portfolios earned more than beta times the realized premium on these days. Neither the Sharpe-Lintner version nor the Black (1972) version, which requires a positive zero-beta rate, accommodates this tilt, and the model's common non-factor term would if anything predict the opposite sign. I disclose it as a characteristic of the amplified state and do not pursue it here.

Finally, the amplification bears on the relation to Andrei, Friedman, and Ozel (2023). Their

mechanism also strengthens with uncertainty, so amplification alone does not distinguish the two channels; what distinguishes them is what conditions the pricing and which direction the mechanism moves posterior factor uncertainty. In their framework, uncertainty draws attention to scheduled announcements whose content loads on the aggregate factor, and the premium is earned as that uncertainty is resolved; consistent with this, their day-level evidence conditions on the attention level and on lagged volatility, and their beta-sorted portfolios show an unconditional steepening of roughly 6 basis points on high-VIX days. My state decomposes that unconditional steepening: mixing the two top-tercile cells at their day counts yields an unconditional high-uncertainty slope of roughly 14 basis points, all of which is carried by the firm-anchored subset and none by the remainder. The mechanisms also leave opposite footprints, which the following robustness exercises examine in detail: the composition premium requires no announcement and survives the excision of earnings windows, does not respond to the supply of news or the level of attention, leaves conditional betas unchanged where the resolution channel raises them, is flat from the first day of an anchored episode through its continuation rather than front-loaded at an information event, and shows no resolution signature, with the change in implied volatility on anchored days indistinguishable from zero once the contemporaneous market return is removed and the premium present on days when uncertainty rises. Attention that resolves factor uncertainty lowers V_f and compresses the premium going forward; attention that crowds out factor learning leaves V_f standing and earns the premium continuously. The amplification documented here is what the second mechanism implies and the first does not.

6.3 Factor specificity and alternative test assets

The mechanism makes two predictions beyond the steepening of the security market line on beta-sorted portfolios. Because attention to firm-specific news moves posterior uncertainty about the market factor and nothing else, the state should change the price of market exposure alone, leaving the premia on size, value, profitability, and investment loadings unchanged across states; and because the premium that emerges on anchored days is compensation for exposure to the factor, market beta should price the cross section on those days regardless of the characteristic on which the test assets are sorted. The first prediction separates the mechanism from stories in which the state moves the price of systematic risk generally, such as time variation in risk aversion or sentiment, which would lift every premium together. The second separates it from the possibility that the state happens to coincide with the pricing of the beta sort alone.

To test the first prediction, I apply the paper's cross-sectional design to each of the five

Fama–French factors in turn. For each factor, every stock's univariate loading is estimated over the trailing 252 trading days and lagged one day, stocks are sorted daily into value-weighted deciles on that loading, and the daily cross-sectional slope of decile returns on decile loadings is averaged within each attention state; a pooled regression with the loading interacted with the firm-anchored indicator provides the formal test of the state difference. Table 4 reports the results. The market row reproduces the headline result on deciles rebuilt from scratch: the price of market beta is 9.4 basis points per day ($t = 3.06$) on anchored days and -2.7 ($t = -0.76$) on unanchored days, a difference of 12.8 basis points ($t = 2.74$). No other loading shows a significant state difference. The value, profitability, and investment loadings are priced in neither state, with state differences of -0.1, -1.2, and 0.4 basis points (t between -0.35 and 0.20). The size loading is priced on anchored days in the univariate design (3.9 basis points, $t = 2.53$), but its state difference is insignificant (4.6, $t = 1.31$), and the anchored-day slope reflects market exposure carried by the size sort rather than a priced size loading: univariate SMB loadings have a cross-sectional correlation of 0.66 with market betas, and once the CAPM beta is held fixed, by regressing individual-stock returns on the beta and the SMB loading orthogonalized against it, the SMB coefficient falls to 1.2 basis points ($t = 1.15$). Panel B gives the time-series counterpart: the market's own premium is 9.30 basis points per day on anchored days against 1.44 on unanchored days ($t = 2.16$ for the difference), while the other four premia show no significant concentration in either state. The state therefore moves the price of aggregate market risk specifically, which is what a mechanism operating on posterior uncertainty about a single market factor predicts, and the three nulls join disagreement and sentiment as predicted zeros that a risk-aversion or broad-attention account would not deliver.

To test the second prediction, Figure 5 pools the test assets of the literature into a single cross section, following Savor and Wilson (2014), who estimate their announcement-day line jointly across beta, size and book-to-market, and industry portfolios. The figure plots forty value-weighted portfolios, ten deciles each sorted on beta, size, book-to-market, and momentum, each with a single full-sample beta so that the two states share the same horizontal positions and differ only in mean returns. Three of the four families are sorted on characteristics unrelated to beta, and two of them, size and momentum, are the sorts on which the unconditional CAPM fails most visibly. On firm-anchored days the forty portfolios line up along a security market line with slope 15.0 basis points per day that passes near the origin, with the size, value, and momentum deciles falling on the same line as the beta deciles rather than above or below it. On unanchored days the same forty portfolios trace a flat line with slope -3.4. On days when institutional attention is anchored on firm-specific news, average returns line up with market beta across whatever the test assets are sorted on, and the only

loading whose price the state changes is the one the model says it should.

7 Additional tests

7.1 A continuous robustness battery

The state cutoffs in Table 2 discard information within each state, and a binary classification cannot say whether the pricing content of the attention residual survives conditioning on the variables that the literature has used to explain time variation in the security market line. Table 5 therefore replaces the state with its underlying continuous signal. The dependent variable is the daily Fama–MacBeth price of beta $\widehat{\lambda}_t$, obtained each day by regressing beta-decile portfolio excess returns on their betas, and the regressor of interest is the five-day rolling median of the attention residual evaluated through day $t-1$ and standardized. The coefficient is the change in the price of beta, in basis points per day, per standard deviation of the smoothed residual, and a negative coefficient indicates a steeper security market line when institutional attention is anchored on firm-specific news. All controls are dated $t-1$ and standardized, and t-statistics use Newey–West standard errors with 21 lags.

The baseline gradient is -9.15 basis points per standard deviation ($t = -2.88$) for value-weighted portfolios and -7.73 ($t = -2.85$) for equal-weighted portfolios. This is the continuous counterpart of the state contrast in Table 2: the anchored and unanchored states sit roughly one and a half standard deviations apart on the smoothed residual, so the gradient reproduces the sixteen-basis-point gap between them without a threshold. Panel A then adds thirteen controls one at a time. The first group captures uncertainty and volatility: the VIX, the option-implied bound on the equity premium SVIX^2 of Martin (2017), realized market variance over the prior 21 days, the variance risk premium, and the cross-sectional dispersion of daily stock returns. The second group captures the level and supply of attention rather than its composition: the aggregate institutional attention index, the firm-news attention index, the share of firms with firm-specific newswire coverage, and the share with an earnings announcement. The third group captures market dynamics: cumulative market returns over the prior five and 21 days and the prior day's price of beta. The last adds dummies for FOMC, CPI, PPI, and payroll announcement days and for the day of the week.

No control moves the gradient by more than a fifth, and most move it by less than five percent. The uncertainty group leaves it within four percent of the baseline, so the pricing content of the residual is not a relabeling of volatility regimes, a conclusion that holds whether volatility is measured by realized variance, by the VIX, by its option-implied premium bound, or by

the gap between the two. The two attention-level controls leave it within seven percent, which restates inside a single regression the result of Section 7.7 that the levels of attention carry no pricing content of their own: what prices beta is the attachment of attention to firm-specific news, not how much attention there is. The lagged price of beta changes the gradient by two percent, so the residual does not merely mark persistent slope regimes, and the announcement calendar changes it by one percent, so the gradient is not an announcement effect in disguise. The only controls that attenuate the gradient at all are return dispersion, by ten percent for value-weighted and eighteen percent for equal-weighted portfolios, and coverage share, by seven and ten percent, and each removes part of the state's own footprint rather than an alternative channel: anchored days are the calm regime in which cross-sectional dispersion is low, and the supply of firm-specific news is the principal determinant of the residual (Table 1). With all controls entered jointly the gradient is -10.34 ($t = -2.13$) for value-weighted and -9.98 ($t = -2.49$) for equal-weighted portfolios, larger in magnitude than the baseline; the wider standard errors reflect the dimensionality of the specification rather than any attenuation of the point estimate.

Panel B adds the two conditioning variables with limited coverage, the daily aggregate disagreement measure of Cookson and Niessner (2023) and the investor sentiment index of Baker and Wurgler (2006), on the 2,477 days through 2021 on which both are available, with a matched baseline so that changes in the coefficient reflect the controls rather than the sample. The gradient on this window is -7.70 ($t = -2.05$) and -7.56 ($t = -2.32$). Disagreement changes it by two to four percent and sentiment by one to eight percent, so neither the disagreement channel of Hong and Sraer (2016) nor a sentiment channel accounts for the pricing content of the residual. The joint specification places twenty regressors on 2,477 observations and leaves the point estimate within ten percent of the window's baseline while its t-statistic, falls below two.

7.2 Expected information consumption and individual firms

A natural concern with the regime variable is that bottom-25% residual days may simply identify days when institutional investors are predictably consuming information at specific firms, in which case the conditional CAPM finding would be subsumed by Ben-Rephael, Carlin, Da, and Israelsen's (2021) Expected Information Consumption (EIC) mechanism. To address this, I construct EIC at the firm-day level following their methodology: a firm receives EIC $= 1$ on day t if it has historically exhibited institutional attention spikes around peer-firm scheduled events, FOMC announcements, or macroeconomic announcements occurring on that day. The construction follows BCDI's specifications closely: I use Fama–French 48-

industry classifications to identify peer firms, require a minimum historical spike rate of 30% on past peer-earnings days for EIC_PEER, and apply BCDI's 50% threshold for FOMC and macroeconomic announcements based on the previous four FOMC days and previous year of macro announcements. The construction produces frequencies and diagnostics in line with BCDI's reported values. The combined EIC = 1 frequency is 5.6% of firm-days, comparable to BCDI's reported 6.7% for EIC_ALL despite the more limited Bloomberg-coverage sample. The AIA spike rate among EIC = 1 firm-days is 30.5%, roughly five times the 5.9% spike rate among EIC = 0 firm-days, closely tracking BCDI's reported differential of 24.2% versus 7.1% in their Table III. Panel A of Table 6 reports the conditional CAPM result on bottom-25% residual days, partitioned by EIC. The headline slope of 20.4 bps ($t = 3.15$) on the full cross-section drops only marginally to 19.1 bps ($t = 3.04$) when I restrict to EIC = 0 firms, which are those that BCDI's framework specifically does not flag as expected to consume information. The conditional CAPM result is therefore not driven by firms that BCDI's mechanism predicts. EIC = 1 firms exhibit particularly strong conditional pricing (44.2 bps, $t = 3.10$), within BCDI's reported range of 8 to 44 bps for EIC = 1 subsamples and consistent with their findings, indicating that the two mechanisms operate complementarily: BCDI's framework identifies firm-day variation in attention-driven pricing, while the regime variable identifies day-level variation in the broader attention environment that affects the entire cross-section.

7.3 Earnings announcements and the conditional CAPM

A natural concern is that the conditional pricing result on bottom-quartile residual-attention days is driven by firms that are simultaneously releasing earnings. Earnings announcements convey systematic information and have been shown to steepen the security market line in their own right (Savor and Wilson, 2016; Chan and Marsh, 2022). If bottom-quartile residual days happen to coincide with earnings seasons, the apparent success of the CAPM on these days could simply reflect the announcement premium documented in prior work rather than a distinct attention-composition channel. To address this concern, I define earnings-window firm-days as firm-days for which the firm has an earnings announcement within five business days before or after the current date, and re-estimate the Fama–MacBeth cross-sectional regressions on bottom-quartile residual-attention days in three subsamples: all firms, firms excluding those in an earnings window, and earnings-window firms only.

Panel B of Table 6 reports the results. On the full cross section, the beta slope is 20.4 basis points ($t = 3.15$) with an insignificant intercept of -2.7 basis points ($t = -0.74$), consistent with the baseline findings. When earnings-window firms are excluded, the beta slope declines

only marginally to 18.8 basis points ($t = 2.98$), remaining highly significant. The intercept is -1.8 basis points ($t = -0.49$), statistically indistinguishable from zero. This subsample retains an average of 1,390 firms per cross section, so the result is not an artifact of a thin sample. Restricting the cross section to earnings-window firms only produces a beta slope of 24.4 basis points ($t = 2.88$), larger in magnitude and consistent with the view that earnings announcements contribute to the pricing of beta through a complementary channel. The intercept for this subsample is 1.4 basis points ($t = 0.22$), again insignificant.

The key finding is that removing all firms near an earnings announcement barely affects the conditional beta slope. The decline from 20.4 to 18.8 basis points is economically small relative to the magnitude of the effect, and the t-statistic remains above conventional thresholds. The attention-composition channel therefore operates across the broad cross section of firms, not only among those releasing earnings. The stronger result within earnings-window firms is consistent with, but not required by, the main finding, and suggests that the two mechanisms operate complementarily: earnings announcements provide a firm-level information event that reinforces the cross-sectional pricing of beta, while the attention-composition state variable identifies a market-wide regime in which that pricing extends to all firms.

7.4 Excluding macroeconomic announcement days and LEADs

A further concern is that the bottom-median residual-attention days may overlap with macroeconomic announcement days (Savor and Wilson, 2014) or with leading earnings announcement days (LEADs) identified by Chan and Marsh (2022). If so, the conditional pricing result could be confounded by the well-documented announcement premium on these days. To address this, I re-estimate the Fama–MacBeth regressions after removing these potentially confounding days from the sample.

Panel C of Table 6 reports the results. The first pair of rows excludes LEAD days, defined as days when influential S&P 500 firms announce earnings, following Chan and Marsh (2022). After removing these days, the value-weighted beta slope is 12.3 basis points ($t = 2.72$) and the equal-weighted slope is 8.5 basis points ($t = 2.13$), both statistically significant. The sample retains 1,343 days. The second pair of rows excludes macroeconomic announcement days, defined as days with scheduled FOMC, employment, or PPI releases following Savor and Wilson (2014). The value-weighted slope is 12.5 basis points ($t = 3.13$) and the equal-weighted slope is 8.6 basis points ($t = 2.38$), again both significant, with 1,554 days remaining. In both specifications, the intercepts are economically small and statistically insignificant, consistent with the predictions of the CAPM.

The magnitudes of the beta slopes are close to unchanged relative to the baseline results in Table 2, and the t-statistics remain well above conventional thresholds. The conditional pricing of beta on bottom-median residual-attention days is therefore not driven by overlap with known macroeconomic or bellwether-earnings announcement days. This finding, combined with the EIC and earnings-window results in Panels A and B of the same table, suggests that the attention-composition channel identifies a broad regime in the information environment that is distinct from the specific event-day channels documented in prior literature.

7.5 Does news carry systematic information?

A final concern is that the news on which attention is anchored, though it survives the market-wide and earnings filters, may still convey information about systematic conditions. If a firm's news moves its beta, the market is learning about systematic conditions from that firm's announcement, in the manner of Patton and Verardo (2012) and the expected-information-consumption channel of Ben-Rephael, Carlin, Da, and Israelsen (2021). Under this reading the security market line would be steep on anchored days because attention resolves factor uncertainty, the mechanism of Andrei, Friedman, and Ozel (2023), rather than because it leaves factor uncertainty unresolved as the model here implies. The two accounts make opposite predictions about which news drives the result: if systematic content is responsible, the pricing should come from firm news that moves beta and not from firm news that does not.

I test this by rebuilding the state from beta-moving and beta-neutral news separately. For each firm I estimate the change in its beta on news days from a regression of its excess return on the market return, a news dummy, and their interaction, and I split firms at the median of this change. I then reconstruct the idiosyncratic-attention index from the news of each half, re-estimate the expanding regression to obtain a new residual, and re-run the cross-sectional test. Panel D of Table 6 reports the results. Rebuilding the state from the half of firms whose news moves beta least prices beta at 5.5 basis points per day on anchored days; rebuilding it from the half whose news moves beta most prices beta at 7.6. The two are of comparable magnitude and the same sign, and the beta-moving half does not price more strongly than the beta-neutral half. Because the systematic-information account requires the pricing to load on beta-moving news, this symmetry is evidence against it. A beta-matched version, which keeps the low-change half within quintiles of firm beta so that the retained firms match the full-sample beta distribution, gives 7.6 basis points, so the pattern is not an artifact of selecting low-beta firms; the composition of the two halves by beta and size is nearly identical.

Two features of the exercise should be read with care. Each half is built from half of the news firms, so its idiosyncratic-attention index correlates 0.78 with the baseline index and its state agrees with the baseline state on 55 percent of days; the price of beta in each half is correspondingly attenuated toward the weaker values reported, and neither is individually significant at conventional levels. The informative comparison is therefore between the two halves rather than the significance of either, and that comparison shows no advantage for the beta-moving news that the systematic-information channel would require.This is consistent with the behavior of betas around firm news more generally. Although betas do rise on news days, by 0.074 ($t = 4.56$), the increase reflects a roughly 40 percent rise in idiosyncratic volatility against only a 5 percent rise in market covariance, so the news on which attention anchors makes firms noisier rather than more systematic, and the conditional pricing of beta is not a relabeling of the resolution of aggregate uncertainty.

7.6 Which firms' news drives the result?

The baseline attention-composition measure constructs the idiosyncratic attention index IAI_t by averaging Bloomberg query scores across all firms with a filtered news article on day t. A natural question is whether the conditional pricing result is driven by news arriving at a specific segment of the cross section. If, for example, only news about high-beta firms or large firms generates the attention state that steepens the SML, the mechanism would be narrower than the model implies. To investigate this, I reconstruct the expanding regression in equation (23) ten times, each time computing IAI_t using only the subset of firms belonging to a single beta decile (or, separately, a single size decile). For each decile-specific residual, I identify bottom-median days and re-estimate the Fama–MacBeth cross-sectional regressions on those days. If the conditional CAPM result appears broadly across decile-specific constructions, then no single part of the firm distribution is responsible for the finding.

Table 10 reports the results when the attention state is defined using news from each beta decile separately. Panel B presents value-weighted portfolio results. Across the ten beta-decile-specific constructions, the beta slope λ_β is positive in every column, ranging from 5.3 basis points (decile 9) to 12.5 basis points (decile 1). The slope is statistically significant at the 5% level or better in six of the ten deciles (deciles 1, 3, 4, 5, 7, and 10), and significant at the 10% level in a seventh (decile 2, $t = 1.85$). The intercepts are uniformly small and insignificant, ranging from -4.4 to 1.4 basis points, none statistically different from zero. The average cross-sectional R^2 is approximately 46% throughout, indicating stable explanatory power regardless of which decile's news is used to construct the state variable. For equal-weighted portfolios in Panel A, the pattern is qualitatively similar though weaker

in magnitude, with the beta slope significant in decile 1 (8.6 basis points, $t = 2.38$) and marginally significant in deciles 3, 4, and 5.

The strongest value-weighted results arise when the attention state is constructed from news about the lowest-beta firms (decile 1: 12.5 basis points, $t = 3.13$) and mid-range beta firms (deciles 3 through 5: 11.0 to 11.5 basis points, t-statistics between 2.7 and 2.9). The weakest results come from deciles 6 and 9. The broad pattern suggests that attention to firm-specific news steepens the SML regardless of where in the beta distribution that news originates, though the effect is somewhat attenuated when the state is defined using news from the highest-beta or highest-attention deciles alone.

Table 11 repeats the exercise using size deciles. Panel B again presents value-weighted results. The beta slope is positive in all ten columns, ranging from 5.7 basis points (decile 8) to 13.0 basis points (decile 7). Significance at the 5% level or better obtains in seven of the ten size deciles (deciles 1, 2, 4, 5, 6, 7, and 10), with an additional decile (decile 9: 11.2 basis points, $t = 2.55$) also significant. The intercepts are again uniformly small and insignificant, consistent with the CAPM. The equal-weighted results in Panel A are weaker in magnitude, with significance at the 5% level in deciles 6 and 7 (7.4 and 7.3 basis points) and marginal significance in decile 5 (6.8 basis points, $t = 1.94$).

The size-decile results indicate that the attention-composition channel is not confined to news about firms of a particular size. News from both small firms (decile 1: 8.2 basis points, $t = 1.98$) and large firms (decile 10: 9.5 basis points, $t = 2.33$) generates attention states in which beta prices the cross section. The strongest results emerge when the state is constructed from mid-to-large capitalization firms (deciles 4 through 7: 10.9 to 13.0 basis points), which may reflect the fact that these firms account for a larger share of aggregate institutional attention and therefore produce more informative variation in the residual.

Taken together, the results in Tables 10 and 11 show that the conditional CAPM finding is not an artifact of news concentrated in any single segment of the cross section. The attention-composition state variable steepens the security market line when constructed from news about low-beta or high-beta firms, small or large firms, and most points in between. This breadth is consistent with the model's prediction that the relevant object is the aggregate share of attention devoted to firm-specific news, not the characteristics of the firms producing that news.

7.7 Alternative measures of attention concentration

The baseline state is built from one particular reduction of the two attention series: the out-of-sample residual of aggregate institutional attention on firm-news attention, estimated with expanding coefficients. None of the results should depend on that choice. Table 7 therefore rebuilds the state from alternative reductions of the same two series, holding every other element of the construction fixed: each candidate series is smoothed by its five-day rolling median, compared with its trailing 252-day quantile, with the comparison direction set so that the state is firm-focused, and the completed indicator is lagged one trading day. Differences across rows are then attributable to the measure, not to the machinery. Three alternatives span the natural possibilities: the share of firm-news attention in aggregate attention, which involves no estimated parameters at all; a residual estimated on a trailing 252-day window, which lets the attention relation drift; and the gap between the trailing percentile ranks of the two series, which is invariant to any monotone transformation of either one.

The composition rows deliver the baseline result throughout. At the median cutoff, the price of beta on firm-anchored days is 8.8 basis points ($t = 2.11$) for the share, 12.9 ($t = 2.79$) for the rolling residual, and 11.7 ($t = 2.42$) for the percentile gap, against 12.6 ($t = 3.00$) for the baseline, with complements flat in every row. At the quartile cutoff every composition measure roughly doubles, to 20.2 ($t = 2.89$), 17.2 ($t = 2.28$), and 28.6 ($t = 2.94$), matching the baseline's 23.1 ($t = 2.85$): deepening the state concentrates the signal, whichever way the state is measured. The share is worth singling out because it is the rawest version, a ratio of the two published series with nothing estimated, and it prices at both cutoffs; the result does not depend on the residualization. Each measure's pricing also lines up with its agreement with the baseline classification (0.83 to 0.85 for the three alternatives), consistent with all of them being noisy readings of one underlying composition state. Two coarser variants, a trailing correlation between the series and a tilt of firm-news relative to macro-topic attention, classify regimes rather than days, agree with the baseline barely above chance, and price weakly; I report them in the Internet Appendix.

The level rows show what the composition rows are not. Classifying days by the raw level of aggregate attention prices nothing: the AIA-high state is 0.4 ($t = 0.07$) at the median cutoff and -15.7 ($t = -1.69$) at the quartile, where its complement prices at 8.9 ($t = 2.64$). Days of elevated firm-news attention fare no better, at 3.1 ($t = 0.63$) and 1.1 ($t = 0.12$). The one level configuration that shows pricing, the complement of AIA-high, is the AIA-low state, and its pricing is inherited rather than its own: aggregate attention is the regressand of the residual, so low-AIA days mechanically overlap the firm-anchored state (68 percent

of them are below-median residual days), and the AIA-low days that are not firm-anchored price beta at -4.2 basis points ($t = -0.67$).

Panel C completes the decomposition. In a pooled regression that lets the two indicators compete, the interaction of beta with the firm-anchored state is 20.0 basis points ($t = 3.06$) while the interaction with the AIA-low state is -4.4 ($t = -0.68$). And the conditioning survives within each half of the attention-level distribution: among low-attention days the firm-anchored interaction is 18.6 ($t = 2.42$) at the day level, and among high-attention days it is positive at the day level and significant in continuous form, with the daily slope loading on the lagged residual at -8.9 basis points per standard deviation ($t = -1.99$). The level of attention neither prices beta on its own nor displaces the composition state when the two compete; the composition of attention conditions the slope at every level of attention, just as it does within every uncertainty regime in Section 6.2. Full two-way cells and estimator detail appear in Internet Appendix.

7.8 Sample splits and the full CRSP universe

Two questions about external validity remain: whether the result is stable over time and whether it is local to the firms on which the attention residual is measured. Table 8 answers both. Panel A splits the state contrast for the firms with institutional attention data by period. The contrast is positive in every cut: 13.0 basis points in the first half of the sample and 18.5 in the second, 12.5 ($t = 2.31$) excluding 2020 and 2022, and 15.2 ($t = 2.84$) excluding 2020 alone, so the result is not a feature of the pandemic or the 2022 tightening cycle. The criterion across subsamples is sign consistency with overlapping confidence intervals, which every cell satisfies; significance at half power is not the standard.

Panel B extends the test assets to the entire market while keeping the state fixed. I form ten daily value-weighted beta deciles from all CRSP common stocks on the NYSE, AMEX, and NASDAQ (6,989 firms and 11.2 million firm-days with a 252-day rolling beta, excluding firm-days with a prior-day price below one dollar) and re-estimate the cross-sectional regressions on the same low-$\widehat{u}$ days. The state contrast is 10.1 basis points ($t = 2.50$) and the bottom-quartile slope 13.0 ($t = 2.77$), and both are positive in each half of the sample and in the crisis-exclusion samples. The pattern does not depend on the weighting or on aggregation into portfolios: on equal-weighted deciles the contrast is 8.9 basis points ($t = 2.33$) and on individual stocks 8.8 ($t = 2.44$), with pooled interactions of 9.1 ($t = 2.45$) and 6.2 ($t = 1.90$). Per unit of beta the slopes are about sixty percent of those in the covered sample, but the full universe spans roughly twice the range of betas, so the return spread across deciles is similar;

the attenuation per unit is what the noisier betas of small stocks imply through the model's measurement channel. Panel C rebuilds the deciles within subsets of the universe. The result does not depend on firm size: the contrast is 10.1 basis points ($t = 2.57$) among stocks below the daily median capitalization and 11.6 ($t = 2.42$) above it, with bottom-quartile slopes of 13.9 and 15.8. Restricting to firms without Bloomberg coverage, so that the test assets share no firm with the sample on which the state is measured, the bottom-quartile slope remains 9.5 basis points ($t = 2.52$) while the below-median contrast is small and insignificant: the deepest anchoring prices beta across the entire market, including firms the attention data never observe, whereas milder anchoring prices it where institutional attention is directed. This is the cross-sectional counterpart of the dose-response pattern in Table 2: the deep state is the broad regime, and the mild state the concentrated one.

7.9 Short-sale constraints

A speculative-beta account in the spirit of Hong and Sraer (2016) would attribute the flat line on other days to optimists bidding up high-beta stocks that pessimists cannot short, so that the flattening should be concentrated among stocks with binding short-sale constraints. Panel A of Table 9 tests this directly. I split firms each day by lagged institutional ownership, the standard proxy for lendable supply (Nagel, 2005), using 13F holdings with a 45-day reporting lag, rebuild the beta deciles within each half, and estimate the state-conditional regressions in each. The other-day slope is -3.5 basis points ($t = -0.98$) among the low-ownership half and -3.2 ($t = -0.89$) among the high-ownership half, with intercepts of 4.8 and 4.7; the extreme terciles give -4.2 and -3.0. The line is equally flat for the stocks that are easiest to short, and the constrained half shows no higher intercept, which is the signature the overpricing account requires. The only difference between the halves is on low-$\widehat{u}$ days, where low-ownership stocks price beta more steeply (10.9 against 6.3 basis points), the side on which a constraint mechanism has no prediction. Together with the absence of a reversal after other days and the insensitivity of the gradient to disagreement in Table 5, this rules out short-sale constraints as the source of the flattening.

7.10 Overnight and intraday returns

Hendershott, Livdan, and Rösch (2020) find that beta is priced overnight, from close to open, and not during trading hours. Because that partition is within the day and mine is across days, the two can be crossed. I decompose each stock's close-to-close return into an intraday component, the open-to-close return, and an overnight component, backed out

from the CRSP total return so that the two compound exactly, form value-weighted beta deciles as before, and estimate the state-conditional regressions on each component. Panel B of Table 9 reports the results. Unconditionally the sample reproduces their finding: the overnight price of beta is 4.7 basis points ($t = 3.68$) and the intraday price -1.1 ($t = -0.55$). Conditioning on the attention state shows that the two partitions do different work. The overnight premium is present in both states, 5.0 basis points on low-$\widehat{u}$ days and 4.5 on other days, and is amplified only in the bottom quartile (9.8, $t = 2.69$). The intraday premium, by contrast, is where the attention state operates: the intraday slope is positive on low-$\widehat{u}$ days, 4.6 basis points below the median and 7.3 in the bottom quartile, and significantly negative on other days, -7.4 ($t = -2.43$), a contrast of about 12 basis points that accounts for essentially all of the close-to-close state contrast on these portfolios. The intraday security market line that is flat on average is therefore the mixture of a priced line on days when attention is anchored on firm-specific news and an inverted line on the rest. The overnight premium these authors document and the premium documented here are complementary: theirs accrues between sessions regardless of attention, and mine accrues during the session when attention is anchored on firms, which is also when the larger intraday declines of other days, the source of the flattening of the close-to-close line, are absent.

8 Betting on and against beta

The betting-against-beta factor of Frazzini and Pedersen (2014) is the traded counterpart of a flat security market line. It is long low-beta and short high-beta stocks with the market exposure hedged, so it earns its premium on days when the line is flat and loses on days when it steepens. If the attention state identifies the days on which beta is priced, it should also identify the days on which the low-beta anomaly is earned, and the price of beta should be predictable enough to trade. This section examines both, using a strategy that requires nothing but the attention state and the beta-sorted portfolios. Each day the strategy holds the spread between the highest- and lowest-beta decile portfolios. On low-$\widehat{u}$ days it is long the spread, betting on beta; on other days it is short the spread, betting against beta. The position for day $t + 1$ is set from attention and news through day t, so the strategy is implementable at the close. Table 12 reports the results.

Among the firms with attention data, the strategy has a Sharpe ratio of 0.80 and a daily CAPM alpha of 10.3 basis points ($t = 3.01$) (Panel A). Its annualized volatility, 31.7%, is identical to that of a strategy that bets on beta every day, which has a Sharpe ratio of 0.40 and a CAPM alpha of -1.2 basis points ($t = -0.44$), so the entire improvement comes from

timing. The alpha is 10.3 basis points ($t = 2.99$) against the Fama–French five factors and 10.6 ($t = 3.03$) when the AQR betting-against-beta factor is added, so the strategy is not a repackaging of either. On all CRSP common stocks the results are essentially the same: a Sharpe ratio of 0.73 and alphas of 10.9 basis points against the CAPM ($t = 2.96$) and 11.4 against the five factors plus BAB ($t = 3.05$).

Panel B locates the betting-against-beta premium. The AQR factor earns 0.51 basis points per day on low-$\widehat{u}$ days and 5.44 on other days, a difference of -4.92 basis points ($t = -1.96$), so that 90% of its cumulative return accrues on the days the attention state classifies as unanchored. Under the bottom-quartile definition the factor loses money on low-$\widehat{u}$ days (-3.39 basis points), the difference is -7.69 ($t = -2.03$), and the other days account for more than all of its return. The anomaly is therefore absent on the days the model predicts a steep, CAPM-consistent line and is earned on the rest: the price of beta and the low-beta premium occupy different days, and the flat unconditional line is the average of the two regimes. I read this as locating the anomaly rather than displacing the funding-constraint account of it. The attention state does not say why low-beta stocks earn a premium on unanchored days; it says that on anchored days they do not.

The beta-neutral benchmark in Panel A shows what kind of premium the state times. A version of the switching strategy built as in Frazzini and Pedersen (2014), which levers each leg to unit ex-ante beta, earns a CAPM alpha of 1.6 basis points ($t = 1.23$) and a five-factor alpha of 1.5 ($t = 1.16$), both insignificant. Under a conditional CAPM a zero-beta portfolio has zero expected excess return in every state, so hedging market exposure removes exactly what the attention state times. The absence of beta-neutral alpha is the result rather than a failure of the strategy. If high-beta stocks were mispriced on anchored days, the mispricing would survive the beta hedge; instead the alpha disappears, so on those days beta is priced correctly and the premium is compensation for market risk.

Panel C shows that the profitability depends on the timeliness of the state. With the classification lagged two days the CAPM alpha falls from 10.3 to 4.9 basis points ($t = 1.41$), and with three days to 2.4. A version that bets on beta only in the most anchored quartile retains 7.3 basis points ($t = 2.30$) at a two-day lag before fading to 4.6 at three days, in line with the attenuation of the price of beta reported in Section 7. The pattern is the same on all CRSP common stocks. All returns are gross of transaction costs and financing, which I do not model; the strategy changes position at each state transition, roughly every one to two weeks, so the exercise measures the economic magnitude of the conditional premium rather than an implementable return.

9 Conclusion

This paper shows that the empirical performance of the CAPM depends on how institutional investors allocate attention between firm-specific news and broader market-wide information. Using firm-specific news from the Dow Jones Institutional Newswire, filtered by a large language model to remove systematic content, and institutional attention data from Bloomberg terminal activity, I construct a daily measure of attention composition: the residual from an expanding regression of aggregate institutional attention on idiosyncratic news attention. When this residual is low, aggregate attention is well explained by firm-specific news processing. When it is high, attention is elevated but directed elsewhere.

On the 52.8% of trading days when the attention-composition residual falls below its trailing median, beta prices the cross section of stock returns. The Fama–MacBeth slope is 12.6 basis points per unit of beta for value-weighted portfolios ($t = 3.00$) and 8.7 for equal-weighted portfolios ($t = 2.21$), rising to 23.1 and 22.6 in the most anchored quartile, with intercepts indistinguishable from zero. On the complementary set of days, the security market line is flat. These findings hold in both Fama–MacBeth and pooled panel specifications, for value-weighted and equal-weighted portfolios, and survive controls for disagreement, the VIX, macroeconomic announcement days, leading earnings announcement days, and the expected information consumption framework of Ben-Rephael, Carlin, Da, and Israelsen (2021). Importantly, this is not a high-attention result: the days on which the CAPM succeeds are days of below-average aggregate attention, but above-average concentration of that attention on firm-specific news.

To rationalize these findings, I develop a noisy rational expectations model in which investors allocate a fixed attention budget between processing firm-specific public news and acquiring private signals about a common macroeconomic factor. Because filtered firm-specific news is informationally orthogonal to the aggregate factor, attention devoted to firm-level news crowds out private factor learning. Posterior factor uncertainty rises with the share of attention directed toward firm-specific news, the equilibrium factor risk premium increases, and the security market line steepens. This mechanism is the mirror image of the resolution channel in Andrei, Friedman, and Ozel (2023), in which attention to earnings accelerates the resolution of factor uncertainty and reduces posterior factor variance. The two channels operate through opposite effects on the same theoretical object and leave opposite footprints: the composition premium requires no announcement, is amplified rather than displaced by prior uncertainty, and is earned where factor uncertainty persists rather than where it is resolved.

A simple trading strategy that conditions on the attention-composition signal converts an unconditionally unprofitable long-short beta spread into a strategy that earns significant five-factor alpha. Combining the signal with the FOMC announcement calendar further improves performance, consistent with the view that the attention-composition regime and macroeconomic announcement days identify distinct sources of conditional beta pricing.

There are several directions for future work. First, the model treats the attention allocation ω as exogenous and does not pin down its cross-day determinants. Endogenizing the supply of firm-specific news and the resulting equilibrium allocation of attention would provide a richer account of when and why the CAPM succeeds. Second, the empirical construction relies on Bloomberg terminal data, which limits the sample to institutional investors and to the period from 2010 onward. Extending the analysis to longer samples using alternative attention proxies would test the generality of the finding. Third, the analysis measures the attention of institutional investors. Whether the composition of retail attention carries pricing content of its own, or instead operates through the price-pressure channel that Da (2025) document for its level, is an open question.

These findings have implications for both researchers and practitioners. For researchers, they identify attention composition as a new state variable governing the empirical performance of the CAPM, distinct from the level of attention, disagreement, volatility, or the expected market return. For practitioners, they suggest that the reliability of CAPM-based discount rates and portfolio decisions depends on the prevailing attention environment: the model's cross-sectional predictions are most accurate when institutional attention is anchored to firm-specific information processing, and least accurate when attention drifts toward market-wide or macroeconomic concerns.

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Table 1: Beta-Decile Characteristics, Attention Regimes, and Determinants of Residual Attention

Panel A: Firm characteristics by beta decile

	Beta decile									
	1 Low	2	3	4	5	6	7	8	9	10 High
Average beta	0.488	0.710	0.838	0.947	1.050	1.149	1.260	1.397	1.582	1.939
Average log market capitalization	14.456	14.636	14.653	14.622	14.574	14.545	14.495	14.432	14.318	14.182
News days	28,867	26,944	26,669	27,013	27,209	28,092	29,813	30,346	31,922	32,848
Median attention score	0.360	0.362	0.364	0.351	0.337	0.336	0.330	0.339	0.353	0.377

Panel B: Regime characteristics across residual-attention quartiles

	Q1 Low	Q2 25–50	Q3 50–75	Q4 High	Q1–Q4 Difference
Firm-news attention, standardized	0.544	−0.217	−0.223	−0.401	0.945
Macro attention, standardized	−0.030	−0.041	0.056	0.154	−0.184
IAI	0.075	0.051	0.050	0.045	0.030
AIA	0.405	0.346	0.397	0.444	−0.039
Firm-news supply	0.063	0.050	0.047	0.042	0.021
VIX	17.465	17.003	17.618	18.248	−0.783
Disagreement	0.249	0.242	0.244	0.250	−0.001
Market return	0.100	0.084	0.079	−0.041	0.141

Panel C: Determinants of residual attention

Continuous drivers (single regressions, drivers at $t-1$)

	Coefficient	t	R^2
Firm-news coverage share	−0.22	−7.28	0.046
Earnings-announcement share	−0.14	−4.77	0.018
Macro-topic attention	+0.06	1.39	0.003
EPU innovation (5-day)	+0.02	1.30	0.000
GPR innovation (5-day)[a]	−0.06	−4.50	0.004

Scheduled announcement days (day-of-week controls; standardized daily objects)

	FOMC	CPI	PPI	Payrolls
Aggregate attention (AIA)	+0.38 (4.77)	+0.11 (1.99)	+0.13 (2.40)	+0.25 (3.96)
Firm-news attention (IAI)	+0.43 (4.66)	−0.02 (−0.31)	−0.06 (−1.02)	+0.06 (1.07)
Firms covered	−0.17 (−2.06)	−0.29 (−4.29)	−0.34 (−4.76)	−0.18 (−2.65)
Attention residual $\hat{u}$	−0.04 (−0.40)	+0.15 (1.98)	+0.27 (3.45)	+0.27 (3.65)
Joint R^2, all drivers and announcement calendar				0.056

Notes: Panel A reports firm characteristics across beta deciles. Average beta is the average estimated 252-day market beta; news days is the number of firm-days with newswire coverage within each decile; attention score is the Bloomberg institutional attention measure. Panel B reports averages across quartiles of residual attention, where Q1 denotes the lowest quartile (attention most anchored on firm-specific news) and Q4 the highest. Firm-news attention and macro attention are the standardized daily firm-news attention index and the standardized macroeconomic attention index of Fisher, Martineau, and Sheng (2022); disagreement is from J. Anthony Cookson's website. Panel C reports determinants of the standardized daily residual. The first block reports single regressions on each standardized driver dated $t-1$ with Newey–West standard errors (21 lags); coverage share is the fraction of panel firms with firm-specific news, earnings share the fraction with an earnings announcement, macro-topic attention the Bloomberg macroeconomic reading index, and EPU and GPR innovations five-day changes in the log daily indices of Baker, Bloom, and Davis (2016) and Caldara and Iacoviello (2022), lagged one trading day. The second block reports coefficients on announcement-day dummies from regressions of each standardized daily object on all four dummies and day-of-week fixed effects, with t-statistics in parentheses.

Table 2: Daily Excess Returns on Low-$\widehat{u}$ and Other Days

	Fama–MacBeth regression				Pooled regression			
Type of day	Intercept	Beta	Avg. R^2	Days	Intercept	Beta	Low $\widehat{u}$	Low $\widehat{u}\times$Beta
Panel A: Ten beta-sorted portfolios (value-weighted)								
Low $\widehat{u}$ days, bottom quartile	−7.00	**23.13**	0.437	601				
	[−1.07]	[**2.85**]						
Low $\widehat{u}$ days	−2.98	**12.63**	0.451	1,668	6.23	−4.42	−10.54	**18.68**
	[−0.91]	[**3.00**]			[1.38]	[−0.96]	[−1.83]	[**3.02**]
Other days	5.34	−3.43	0.450	1,493				
	[1.37]	[−0.71]						
Low $\widehat{u}$ − Other	−8.32	**16.06**						
	[−1.64]	[**2.53**]						
Panel B: Ten beta-sorted portfolios (equal-weighted)								
Low $\widehat{u}$ days	2.36	**8.73**	0.551	1,668	6.36	−4.32	−4.13	**13.36**
	[0.94]	[**2.21**]			[1.47]	[−1.00]	[−0.76]	[**2.31**]
Other days	6.32	−4.20	0.567	1,493				
	[2.02]	[−0.93]						
Low $\widehat{u}$ − Other	−3.95	**12.93**						
	[−0.99]	[**2.17**]						
Panel C: Individual stocks								
Low $\widehat{u}$ days	4.33	**6.46**	0.039	1,668	−0.02	1.44	−0.26	**9.27**
	[2.36]	[**2.00**]			[−0.01]	[0.56]	[−0.04]	[**1.87**]
Other days	4.84	−2.96	0.047	1,493				
	[2.13]	[−0.75]						
Low $\widehat{u}$ − Other	−0.51	**9.42**						
	[−0.17]	[**1.85**]						

Notes: Fama–MacBeth regressions of daily excess returns on betas, estimated separately on low-$\widehat{u}$ days, when institutional attention is anchored on firm-specific news, and on other days, with a pooled regression adding a low-$\widehat{u}$ dummy and its interaction with beta. Day t is a low-$\widehat{u}$ day when the five-day rolling median of $\widehat{u}$ through $t-1$ lies below its trailing 252-day median; the bottom-quartile row uses the 25th percentile. The test assets are formed from the firms with Bloomberg institutional attention data, on which $\widehat{u}$ is measured; Table 8 repeats the tests on all CRSP common stocks. Panels A and B use ten beta-sorted portfolios formed daily on lagged 252-day rolling betas, value- and equal-weighted, with the decile's average beta as regressor; Panel C uses individual stocks. Coefficients are in basis points per day. Fama–MacBeth t-statistics (brackets) use Newey–West standard errors with five lags; the difference row regresses the daily coefficients on the low-$\widehat{u}$ indicator. Pooled t-statistics are clustered by trading day.

Table 3: Uncertainty and the Composition Premium

Panel A: Price of beta by prior-day uncertainty tercile				
	Firm-anchored state			Unanchored
	T1 Low	T2	T3 High	T3 Placebo
Realized variance	−0.4	14.7***	24.2**	−8.6
	(−0.08)	(2.69)	(2.14)	(−0.82)
VIX	6.9*	1.4	31.8***	−2.8
	(1.84)	(0.26)	(2.92)	(−0.25)
JLN $h = 1$	12.8***	1.7	24.1**	−0.4
	(2.95)	(0.30)	(2.34)	(−0.03)
JLN $h = 12$	10.8**	3.5	24.9**	−0.3
	(2.51)	(0.63)	(2.32)	(−0.03)

Panel B: Pooled interaction within the top uncertainty tercile			
	β	$\beta \times D^{state}$	Days
Realized variance	−14.1	33.8**	1,054
	(−1.40)	(2.33)	
VIX	−7.0	30.3**	1,054
	(−0.65)	(2.07)	
JLN $h = 1$	−9.1	26.1*	1,046
	(−0.86)	(1.80)	
JLN $h = 12$	−7.1	24.4*	1,047
	(−0.73)	(1.77)	

Panel C: Variance risk premium within the anchored high-uncertainty cell			
	λ_β	Intercept	Days
Low variance risk premium	10.4	1.7	261
	(0.63)	(0.14)	
High variance risk premium	51.8***	−28.6**	261
	(3.60)	(−2.21)	

Notes: Panel A reports Fama–MacBeth prices of beta in basis points per day on firm-anchored days, sorted into terciles of the prior trading day's uncertainty proxy, with breakpoints computed within sample; the final column repeats the top-tercile estimation on unanchored days. The attention state is the five-day rolling median of the residual $\widehat{u}$ relative to its trailing 252-day median, lagged one trading day, so all conditioning information precedes the return day. Anchored cells contain 495 to 623 days and span more than thirty calendar quarters, with no quarter contributing more than ten percent. Panel B estimates $r_{p,t} = \alpha + \beta\,\widehat{\beta}_p + \gamma D_t^{state} + \delta(\widehat{\beta}_p \times D_t^{state}) + \varepsilon_{p,t}$ on top-tercile days only, with standard errors clustered by trading day. Panel C sorts the anchored top-SVIX^2-tercile days by the variance risk premium, SVIX^2 minus trailing 21-day realized variance, where SVIX^2 is the option-implied lower bound on the conditional equity premium of Martin (2017); the attention residual is uncorrelated with SVIX^2 (correlation 0.009). JLN denotes the macroeconomic uncertainty index of Jurado, Ludvigson, and Ng (2015), mapped by prior calendar month. Newey–West t-statistics with five lags are in parentheses in Panels A and C. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

Table 4: Factor Specificity: The Price of Factor Exposures and Factor Returns by Attention State

Panel A: Fama–MacBeth price of each factor loading, univariate loadings			
Factor	Low $\widehat{u}$ days	Other days	Difference (pooled)
Market	9.4***	−2.7	12.8***
	(3.06)	(−0.76)	(2.74)
SMB	3.9**	−1.5	4.6
	(2.53)	(−0.87)	(1.31)
HML	0.9	−0.1	−0.1
	(0.50)	(−0.05)	(−0.02)
RMW	−0.5	0.3	−1.2
	(−0.37)	(0.20)	(−0.35)
CMA	−1.7*	−0.1	0.4
	(−1.67)	(−0.08)	(0.20)
Panel B: Fama–MacBeth price of each factor loading, multivariate (partial) loadings			
Factor	Low $\widehat{u}$ days	Other days	Difference (pooled)
Market	9.8***	−2.2	13.3***
	(2.96)	(−0.57)	(2.64)
SMB	2.6*	−1.6	3.0
	(1.82)	(−0.96)	(1.16)
HML	1.0	−2.1	3.2
	(0.59)	(−1.17)	(1.33)
RMW	0.3	1.3	−1.2
	(0.27)	(1.11)	(−0.70)
CMA	−3.0***	0.4	−2.0
	(−2.84)	(0.38)	(−1.30)
Days	1,668	1,493	
Panel C: Mean daily factor return by attention state			
Factor	Low $\widehat{u}$ days	Other days	Difference t
Market	9.30	1.44	2.16**
SMB	1.07	−2.42	1.59
HML	−0.79	−0.05	−0.25
RMW	0.38	1.76	−0.79
CMA	−0.74	0.39	−0.72

Notes: Panels A and B apply the paper's cross-sectional design to each Fama–French factor in turn. In Panel A each stock's loading on a factor is estimated from a univariate 252-day rolling regression of its excess return on that factor; in Panel B the five loadings are estimated jointly from a 252-day rolling regression on all five factors, so each is a partial loading. Loadings are lagged one day. For each factor, stocks are sorted daily into deciles on their loading, decile excess returns are value-weighted using prior-day market capitalization, and the decile's average loading is the regressor. The first two columns report time-series means of the daily cross-sectional slopes on low-$\widehat{u}$ days and on other days, with Newey–West t-statistics (five lags); the third reports the coefficient on the loading interacted with the low-$\widehat{u}$ indicator in a pooled regression of decile returns on the loading, the indicator, and their interaction, with standard errors clustered by day. The average daily cross-sectional correlation between the SMB loading and the market loading is 0.66 for univariate loadings and 0.19 for partial loadings; the HML, RMW, and CMA partial loadings correlate with the market loading at 0.09, −0.06, and −0.03. Panel C reports the mean daily return of each factor on low-$\widehat{u}$ days and on other days, and the t-statistic on the state indicator in a regression of the factor return on that indicator (Newey–West, five lags). Coefficients are in basis points per day. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

Table 5: Continuous Robustness of the Attention Gradient in the Price of Beta

Panel A: Full sample

	Value-weighted			Equal-weighted			
Specification	b	(t)	R^2	b	(t)	R^2	Days
Baseline	-9.15^{***}	(-2.88)	0.002	-7.73^{***}	(-2.85)	0.002	3,220
+ VIX	-9.30^{***}	(-2.92)	0.004	-7.89^{***}	(-2.93)	0.005	3,220
+ SVIX^2	-9.53^{***}	(-2.98)	0.005	-8.08^{***}	(-2.98)	0.005	3,220
+ Realized variance (21 days)	-9.24^{***}	(-2.90)	0.002	-7.85^{***}	(-2.89)	0.002	3,220
+ Variance risk premium	-9.11^{***}	(-2.87)	0.002	-7.72^{***}	(-2.84)	0.002	3,220
+ Return dispersion	-8.27^{***}	(-2.59)	0.003	-6.36^{**}	(-2.38)	0.006	3,220
+ Aggregate attention (AIA)	-8.53^{***}	(-2.65)	0.002	-7.82^{***}	(-2.82)	0.002	3,220
+ Firm-news attention (IAI)	-9.24^{***}	(-2.86)	0.002	-7.75^{***}	(-2.83)	0.002	3,220
+ Firm-news coverage share	-8.50^{**}	(-2.43)	0.002	-6.93^{**}	(-2.34)	0.002	3,220
+ Earnings-announcement share	-9.02^{***}	(-2.73)	0.002	-7.52^{***}	(-2.66)	0.002	3,220
+ Lagged 5-day market return	-10.22^{***}	(-3.02)	0.008	-8.58^{***}	(-3.01)	0.006	3,220
+ Lagged 21-day market return	-9.60^{***}	(-2.97)	0.003	-7.99^{***}	(-2.93)	0.002	3,220
+ Lagged price of beta	-8.94^{***}	(-2.90)	0.003	-7.32^{***}	(-2.85)	0.004	3,219
+ Announcement and day-of-week dummies	-9.09^{***}	(-2.89)	0.006	-7.59^{***}	(-2.83)	0.005	3,220
+ All controls	-10.34^{**}	(-2.13)	0.017	-9.98^{**}	(-2.49)	0.017	3,219

Panel B: Common sample with disagreement and sentiment

	Value-weighted			Equal-weighted			
Specification	b	(t)	R^2	b	(t)	R^2	Days
Baseline	-7.70^{**}	(-2.05)	0.002	-7.56^{**}	(-2.32)	0.002	2,477
+ Disagreement	-7.57^{**}	(-1.96)	0.002	-7.26^{**}	(-2.19)	0.002	2,477
+ Sentiment	-7.05^{**}	(-1.97)	0.002	-7.46^{**}	(-2.36)	0.002	2,477
+ Disagreement and sentiment	-7.09^{*}	(-1.93)	0.002	-7.26^{**}	(-2.25)	0.002	2,477
+ All controls	-6.91	(-1.23)	0.021	-7.20	(-1.61)	0.020	2,476

Notes: The table reports time-series regressions of the daily Fama–MacBeth price of beta $\widehat{\lambda}_t$, in basis points per day, on the standardized five-day rolling median of the attention residual $\widehat{u}$ evaluated through day $t-1$, with controls added as indicated. Each day, beta-decile portfolio excess returns are regressed cross-sectionally on their betas to obtain $\widehat{\lambda}_t$; the left block uses value-weighted and the right block equal-weighted portfolios. The reported coefficient b is in basis points per one-standard-deviation increase in the smoothed residual, so a negative value indicates a steeper security market line when institutional attention is anchored on firm-specific news. All controls are standardized and dated $t-1$: VIX; SVIX^2, the option-implied bound on the equity premium of Martin (2017); realized variance, the sum of squared daily market excess returns over the prior 21 trading days; the variance risk premium, SVIX^2 minus annualized realized variance; return dispersion, the cross-sectional standard deviation of daily stock returns; AIA and IAI, the aggregate institutional attention and firm-news attention indices; coverage share, the fraction of panel firms with firm-specific newswire coverage; earnings-announcement share, the fraction with an earnings announcement; cumulative market excess returns over the prior 5 and 21 trading days; the prior day's price of beta; and dummies for FOMC, CPI, PPI, and payroll announcement days and for day of the week. The all-controls specification includes every control except SVIX^2, whose correlation with VIX is 0.998. Panel B restricts the sample to days on which the disagreement measure of J. Anthony Cookson and the sentiment index of Baker and Wurgler (2006) (orthogonalized to macroeconomic conditions and mapped by prior calendar month) are both available, and reports a matched baseline so that changes in the coefficient reflect the controls rather than the sample; the all-controls row in Panel B adds twenty regressors to 2,477 observations, and its point estimate rather than its t-statistic is the informative quantity. Newey–West t-statistics with 21 lags are in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

Table 6: Robustness to Information-Event Channels

Panel A: Expected information consumption (Ben-Rephael, Carlin, Da, and Israelsen, 2021)

	All firms	EIC = 0 firms	EIC = 1 firms
α	−2.72	−2.12	−18.77
	(−0.74)	(−0.60)	(−1.54)
λ_β	20.44***	19.11***	44.16***
	(3.15)	(3.04)	(3.10)
Mean firms per day	1,750	1,635	168
Days	449	449	282

Panel B: Earnings announcement windows (Savor and Wilson, 2016)

	All firms	Excl. earnings window	Earnings window only
α	−2.72	−1.75	1.36
	(−0.74)	(−0.49)	(0.22)
λ_β	20.44***	18.81***	24.43***
	(3.15)	(2.98)	(2.88)
Mean firms per day	1,750	1,390	396
Days	449	449	403

Panel C: Macro-news days (Chan and Marsh, 2022; Savor and Wilson, 2013)

	α	λ_β	Days
Excluding LEAD days, EW	0.55	8.52**	1343
	(0.20)	(2.13)	
Excluding LEAD days, VW	−5.25	12.33***	1343
	(−1.39)	(2.72)	
Excluding macro-announcement days, EW	0.73	8.58**	1554
	(0.30)	(2.38)	
Excluding macro-announcement days, VW	−4.87	12.47***	1554
	(−1.48)	(3.13)	

Panel D: State rebuilt from beta-moving versus beta-neutral news

	λ_β (anchored)	λ_β (other)	Firms
All firm news (baseline)	12.63***	−3.43	1,627
	(3.00)	(−0.71)	
Beta-neutral news only	5.50	4.65	814
	(1.14)	(1.03)	
Beta-moving news only	7.62*	2.79	813
	(1.70)	(0.63)	
Beta-neutral, beta-matched	7.63*	2.71	815
	(1.72)	(0.58)	

Notes: Fama–MacBeth regressions of excess returns on CAPM betas (estimated over the previous 252 days); Newey–West t-statistics in parentheses (10 lags in Panels A–B, 5 in C–D). Coefficients are in basis points per day. Panels A and B use individual firms on bottom-quartile residual-attention days. In Panel A, EIC = 1 firms are those whose institutional attention is predicted to spike around peer-firm, FOMC, and macroeconomic announcements, following Ben-Rephael, Carlin, Da, and Israelsen (2021); EIC = 0 firms are the rest. In Panel B, earnings-window firm-days are those within five business days of an earnings announcement. Panel C uses beta-decile portfolios on below-median residual-attention days, excluding LEAD days (Chan and Marsh) or macro-announcement days (Savor and Wilson); equal-weighted (EW) and value-weighted (VW). Panel D rebuilds the attention residual from a subset of firms and reports the value-weighted price of beta by state. Each firm's news-day beta change is estimated from a regression of its excess return on the market return, a news dummy, and their interaction (Andrei, Friedman, and Ozel, 2023); firms are split at the median. "Beta-neutral" and "beta-moving" rebuild the residual from the halves whose news moves beta least and most; "beta-matched" keeps the low-change half within firm-beta quintiles. Sample 2011–2025. ***, **, *: 1%, 5%, 10% significance.

Table 7: Alternative Measures of Attention Concentration

	Firm-anchored days				Complement		
Measure	λ_β (VW)	λ_β (EW)	α (VW)	Days	λ_β (VW)	Days	Agreement
Panel A: Median cutoff							
Expanding residual (baseline)	12.6***	8.7**	−3.0	1,668	−3.4	1,493	1.00
	(3.00)	(2.21)	(−0.91)		(−0.71)		
Share IAI/AIA	8.8**	6.1	0.1	1,791	−1.0	1,430	0.85
	(2.11)	(1.55)	(0.02)		(−0.22)		
Rolling-252 residual	12.9***	8.4*	−4.7	1,402	−2.2	1,567	0.83
	(2.79)	(1.92)	(−1.28)		(−0.48)		
Percentile gap	11.7**	7.9*	−2.8	1,469	−1.8	1,501	0.85
	(2.42)	(1.75)	(−0.70)		(−0.37)		
Level: AIA high	0.4	0.1	4.4	1,486	8.0**	1,735	0.33
	(0.07)	(0.02)	(1.06)		(2.24)		
Level: IAI high	3.1	3.6	3.5	1,637	5.9*	1,584	0.51
	(0.63)	(0.80)	(0.88)		(1.65)		
Panel B: Bottom-quartile cutoff							
Expanding residual (baseline)	23.1***	22.6***	−7.0	601	0.8	2,560	
	(2.85)	(3.07)	(−1.07)		(0.23)		
Share IAI/AIA	20.2***	18.2***	−6.8	673	0.3	2,548	
	(2.89)	(2.98)	(−1.19)		(0.09)		
Rolling-252 residual	17.2**	14.2**	−6.3	567	2.0	2,402	
	(2.28)	(1.98)	(−1.03)		(0.53)		
Percentile gap	28.6***	23.0***	−15.3*	530	−0.2	2,440	
	(2.94)	(2.58)	(−1.96)		(−0.07)		
Level: AIA high	−15.7*	−13.4*	17.4**	577	8.9***	2,644	
	(−1.69)	(−1.67)	(2.28)		(2.64)		
Level: IAI high	1.1	2.6	2.6	610	5.3*	2,611	
	(0.12)	(0.28)	(0.31)		(1.70)		

Panel C: Attention level versus attention composition

	Joint pooled regression, full panel (day-clustered)			
	$\beta\times$ firm-anchored		$\beta\times$ AIA-low	
	20.0*** (3.06)		−4.4 (−0.68)	
	Composition interactions within attention-level halves			
	$\beta\times$ anchored (day-level)	$\beta\times$ anchored (pooled)	$\beta \times z(\hat{u})$	Days
Within AIA-low	18.6** (2.42)	16.2* (1.91)	−5.4 (−1.62)	1,726
Within AIA-high	11.5 (0.97)	24.0** (2.40)	−8.9** (−1.99)	1,435

Notes: Each measure is pushed through the baseline state construction: five-day rolling median compared with the trailing 252-day quantile (median in Panel A, quartile in Panel B, direction set so the state is firm-focused), lagged one trading day. Fama–MacBeth prices of beta in basis points per day; Newey–West t-statistics (five lags) in parentheses. Agreement is the share of days classified identically to the baseline. The complement of the AIA-high rows is the AIA-low state; its pricing reflects mechanical overlap with the residual state, of which AIA is the regressand: AIA-low days that are not firm-anchored price beta at −4.2 ($t = -0.67$). Panel C interacts beta with the firm-anchored and AIA-low indicators jointly (day-clustered), and, within each half of the attention-level distribution, reports the firm-anchored interaction from daily slopes regressed on the state (day-level, equal-weighted across days) and from the pooled panel (dispersion-weighted), plus the interaction with the standardized lagged residual $z(\hat{u})$. Day-level estimates equal differences of the corresponding Fama–MacBeth cells. ***, **, *: 1%, 5%, 10%.

Table 8: Sample Splits and the Full CRSP Universe

Panel A: Firms with institutional attention data, by period

	Low $\widehat{u}$ − Other		
Sample	Coef.	(t)	Days
Full sample	16.1	(2.53)	3,161
Pre-2018	13.0	(1.89)	1,410
Post-2018	18.5	(1.85)	1,751
Excluding 2020 and 2022	12.5	(2.31)	2,657
Excluding 2020	15.2	(2.84)	2,908

Panel B: All CRSP common stocks, by period

	Fama–MacBeth slope		Low $\widehat{u}$ − Other		Bottom quartile	
Sample	Low $\widehat{u}$	Other	Coef.	(t)	Coef.	(t)
Full sample	7.6	−2.6	10.1	(2.50)	13.0	(2.77)
Pre-2018	4.8	−3.1	8.0	(1.92)	9.3	(1.88)
Post-2018	9.8	−2.1	11.9	(1.82)	15.4	(2.20)
Excluding 2020 and 2022	7.4	−0.7	8.2	(2.37)	5.8	(1.41)
Excluding 2020	7.2	−3.3	10.5	(2.94)	8.2	(1.96)

Panel C: All CRSP common stocks, by subset of firms

	Fama–MacBeth slope		Low $\widehat{u}$ − Other		Bottom quartile	
Subset	Low $\widehat{u}$	Other	Coef.	(t)	Coef.	(t)
Large (above median capitalization)	8.9	−2.7	11.6	(2.42)	15.8	(2.82)
Small (below median capitalization)	5.4	−4.7	10.1	(2.57)	13.9	(3.04)
Firms with Bloomberg coverage	9.6	−2.8	12.4	(2.65)	16.5	(2.91)
Firms without Bloomberg coverage	2.8	−0.6	3.4	(0.96)	9.5	(2.52)

Notes: Panel A reports, for the firms with institutional attention data, the difference between the mean daily value-weighted Fama–MacBeth price of beta on low-$\widehat{u}$ days and on other days within each subperiod. Panels B and C use ten daily value-weighted beta deciles formed from all CRSP common stocks on NYSE, AMEX, and NASDAQ with a 252-day rolling beta and a prior-day price of at least one dollar, with the same attention state as Table 2; Panel C rebuilds the deciles within each subset of firms. Coefficients are in basis points per day; Newey–West t-statistics with five lags in parentheses. The bottom-quartile column is the Fama–MacBeth slope on days below the trailing 25th percentile of $\widehat{u}$.

Table 9: Short-Sale Constraints and the Intraday Decomposition of the Price of Beta

Panel A: Price of beta by institutional ownership

	Low $\widehat{u}$ days		Other days		Low $\widehat{u}$ − Other	
Institutional ownership	Beta	Intercept	Beta	Intercept	Coef.	(t)
Below median (constrained)	10.9	−1.1	−3.5	4.8	14.4	(3.04)
	(3.49)		(−0.98)			
Above median (unconstrained)	6.3	2.6	−3.2	4.7	9.5	(2.02)
	(2.08)		(−0.89)			
Lowest tercile	11.5	−1.8	−4.2	5.5	15.7	(3.30)
	(3.70)		(−1.17)			
Highest tercile	5.5	3.7	−3.0	3.8	8.5	(1.75)
	(1.78)		(−0.80)			

Panel B: Overnight and intraday returns

		Intercept		Beta	
Return component	Type of day	Coef.	(t)	Coef.	(t)
Close to close	All days	2.50	(1.37)	3.78	(1.53)
	Low $\widehat{u}$, bottom quartile	−0.74	(−0.17)	16.94	(2.89)
	Low $\widehat{u}$, below median	0.33	(0.14)	9.62	(3.00)
	Other days	4.92	(1.72)	−2.74	(−0.71)
Overnight	All days	−0.38	(−0.42)	4.73	(3.68)
	Low $\widehat{u}$, bottom quartile	−1.67	(−0.69)	9.75	(2.69)
	Low $\widehat{u}$, below median	−0.59	(−0.52)	4.97	(3.05)
	Other days	−0.15	(−0.10)	4.47	(2.08)
Intraday	All days	2.99	(1.84)	−1.06	(−0.55)
	Low $\widehat{u}$, bottom quartile	0.95	(0.25)	7.25	(1.70)
	Low $\widehat{u}$, below median	0.99	(0.47)	4.60	(1.87)
	Other days	5.22	(2.03)	−7.37	(−2.43)

Notes: Both panels use the firms with institutional attention data. In Panel A, firms are sorted each day by institutional ownership from 13F filings, lagged 45 days after the quarter-end, and ten value-weighted beta deciles are rebuilt within each group; the panel reports the mean daily Fama–MacBeth slope and intercept of decile excess returns on lagged 252-day rolling betas on low-$\widehat{u}$ days and on other days, and the difference in slopes. Low institutional ownership proxies for limited lendable supply and hence binding short-sale constraints (Nagel, 2005). In Panel B, following Hendershott, Livdan, and Rösch (2020), each stock's daily close-to-close return is decomposed into an intraday component, the open-to-close return from CRSP open and close prices, and an overnight component backed out from the CRSP total return so that the two compound to it exactly; ten value-weighted beta deciles are formed daily on lagged 252-day rolling betas and each component's decile returns are regressed on the decile's average beta. The panel reports time-series means of the daily intercepts and slopes on all days, on low-$\widehat{u}$ days below the trailing median and below the trailing 25th percentile of $\widehat{u}$, and on other days (3,161, 1,668, 601, and 1,493 days). Coefficients are in basis points per period; Newey–West t-statistics with five lags in parentheses.

Table 10: Fama–MacBeth Regressions: Attention Explained by Firm News Across Beta Deciles

	Beta decile used to define the attention-state condition									
	1	2	3	4	5	6	7	8	9	10
Panel A: Equal-weighted returns										
Intercept	1.23	2.84	1.47	2.37	1.87	5.47**	2.00	3.29	5.39**	3.47
	(0.51)	(1.10)	(0.60)	(0.99)	(0.71)	(2.25)	(0.83)	(1.33)	(2.05)	(1.49)
λ_β	8.58**	4.50	6.31*	5.84*	6.61*	2.09	3.98	4.68	1.61	5.14
	(2.38)	(1.20)	(1.75)	(1.66)	(1.73)	(0.56)	(1.12)	(1.26)	(0.41)	(1.46)
Avg. R^2	0.547	0.562	0.555	0.558	0.548	0.560	0.557	0.557	0.551	0.558
Days	1,554	1,461	1,415	1,482	1,437	1,500	1,531	1,568	1,528	1,635
Panel B: Value-weighted returns										
Intercept	−4.37	−0.54	−4.10	−4.24	−3.96	0.92	−1.77	−0.87	1.43	0.06
	(−1.33)	(−0.16)	(−1.29)	(−1.42)	(−1.15)	(0.29)	(−0.57)	(−0.27)	(0.42)	(0.02)
λ_β	12.47***	7.62*	11.47***	11.01***	11.48***	6.11	7.51**	8.40**	5.31	8.52**
	(3.13)	(1.85)	(2.91)	(2.93)	(2.72)	(1.52)	(1.99)	(2.08)	(1.27)	(2.22)
Avg. R^2	0.454	0.471	0.461	0.461	0.459	0.459	0.462	0.463	0.462	0.464
Days	1,554	1,461	1,415	1,482	1,437	1,500	1,531	1,568	1,528	1,635

Notes: This table reports Fama–MacBeth regressions of daily beta-decile portfolio returns on portfolio betas, restricted to days on which aggregate attention is well explained by firm-specific news among firms in the indicated beta decile. Each day, portfolio returns are regressed cross-sectionally on beta, and the table reports time-series averages of the daily intercepts and slopes in basis points per day, with Newey–West t-statistics (five lags) in parentheses. Avg. R^2 is the average daily cross-sectional R^2. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

Table 11: Fama–MacBeth Regressions: Attention Explained by Firm News Across Size Deciles

	Size decile used to define the attention-state condition									
	1 Small	2	3	4	5	6	7	8	9	10 Large
Panel A: Equal-weighted returns										
Intercept	4.32*	3.93	4.38*	3.16	3.43	3.22	3.09	3.89	2.28	1.79
	(1.66)	(1.63)	(1.85)	(1.24)	(1.43)	(1.32)	(1.29)	(1.54)	(0.84)	(0.71)
λ_β	3.23	4.16	2.54	5.68	6.82*	7.42**	7.33**	1.70	6.28	5.11
	(0.86)	(1.16)	(0.69)	(1.52)	(1.94)	(2.03)	(2.08)	(0.44)	(1.54)	(1.34)
Avg. R^2	0.553	0.558	0.557	0.558	0.553	0.553	0.549	0.555	0.555	0.558
Days	1,495	1,526	1,479	1,529	1,532	1,492	1,521	1,472	1,534	1,489
Panel B: Value-weighted returns										
Intercept	−0.71	−0.80	0.71	−2.90	−1.66	−2.22	−3.26	0.29	−3.39	−2.80
	(−0.21)	(−0.25)	(0.22)	(−0.89)	(−0.54)	(−0.69)	(−1.04)	(0.09)	(−0.97)	(−0.85)
λ_β	8.19**	8.67**	6.10	10.90***	10.92***	11.61***	12.96***	5.67	11.18**	9.53**
	(1.98)	(2.18)	(1.48)	(2.67)	(2.88)	(2.89)	(3.36)	(1.37)	(2.55)	(2.33)
Avg. R^2	0.461	0.461	0.463	0.464	0.463	0.462	0.452	0.468	0.459	0.464
Days	1,495	1,526	1,479	1,529	1,532	1,492	1,521	1,472	1,534	1,489

Notes: This table reports Fama–MacBeth regressions of daily beta-decile portfolio returns on portfolio betas, restricted to days on which aggregate attention is well explained by firm-specific news among firms in the indicated size decile; decile 1 contains the smallest firms and decile 10 the largest. Each day, portfolio returns are regressed cross-sectionally on beta, and the table reports time-series averages of the daily intercepts and slopes in basis points per day, with Newey–West t-statistics (five lags) in parentheses. Avg. R^2 is the average daily cross-sectional R^2. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

Table 12: State-Switching Beta Strategies

Panel A: Strategy performance						
	Ann. return	Ann. vol.	Sharpe	CAPM α	FF5 α	FF5+BAB α
State-switching strategy						
Firms with attention data	25.2%	31.7%	0.80	10.3***	10.3***	10.6***
				(3.01)	(2.99)	(3.03)
All CRSP common stocks	25.0%	34.1%	0.73	10.9***	10.9***	11.4***
				(2.96)	(2.95)	(3.05)
Benchmarks (firms with attention data)						
Always bet on beta	12.6%	31.7%	0.40	−1.2		
				(−0.44)		
State-switching, beta-neutral	4.6%	11.2%	0.41	1.6	1.5	
				(1.23)	(1.16)	

Panel B: The AQR betting-against-beta factor by attention state					
State definition	Low-$\widehat{u}$ days	Other days	Difference	(t)	Share of return on other days
Median	0.51	5.44***	−4.92*	(−1.96)	90%
Bottom quartile	−3.39	4.30***	−7.69**	(−2.03)	123%

Panel C: CAPM α by age of the state classification					
	Classification lag k (days)				
	1	2	3	4	5
Attention data, median state	10.3	4.9	2.4	3.1	4.1
	(3.01)	(1.41)	(0.69)	(0.86)	(1.14)
Attention data, bottom quartile	8.3	7.3	4.6	3.9	3.1
	(2.51)	(2.30)	(1.48)	(1.15)	(0.96)
All CRSP, median state	10.9	4.8	2.9	5.2	6.0
	(2.96)	(1.28)	(0.76)	(1.36)	(1.59)
All CRSP, bottom quartile	9.5	8.9	5.6	5.8	4.7
	(2.82)	(2.70)	(1.70)	(1.61)	(1.42)

Notes: The state-switching strategy holds the highest-minus-lowest beta decile portfolio (value-weighted, betas from lagged 252-day rolling regressions) on low-$\widehat{u}$ days and the reverse position on other days. Day $t+1$ is a low-$\widehat{u}$ day when the five-day rolling median of $\widehat{u}$ through day t lies below its trailing 252-day median (1,668 of 3,161 days). Panel A reports annualized returns, volatilities, and Sharpe ratios, and daily alphas in basis points from regressions of strategy returns on the market factor, the Fama–French five factors, and the five factors plus the AQR betting-against-beta (BAB) factor, all measured on the holding day. The always-bet-on-beta benchmark holds the same spread every day. The beta-neutral strategy switches between the rank-weighted low- and high-beta portfolios of Frazzini and Pedersen (2014), each levered to unit ex-ante beta. Panel B reports the mean daily return of the AQR U.S. BAB factor on low-$\widehat{u}$ days and other days, under the median and the bottom-quartile (trailing 25th percentile) definitions, the difference, and the share of its cumulative return earned on other days. Panel C repeats the CAPM alpha of the state-switching strategy with the state classified k days before the holding day; $k=1$ is the baseline, and the bottom-quartile rows use the trailing 25th percentile. Returns are gross of transaction costs. Newey–West t-statistics with five lags in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels. Sample: May 2012 to December 2024.

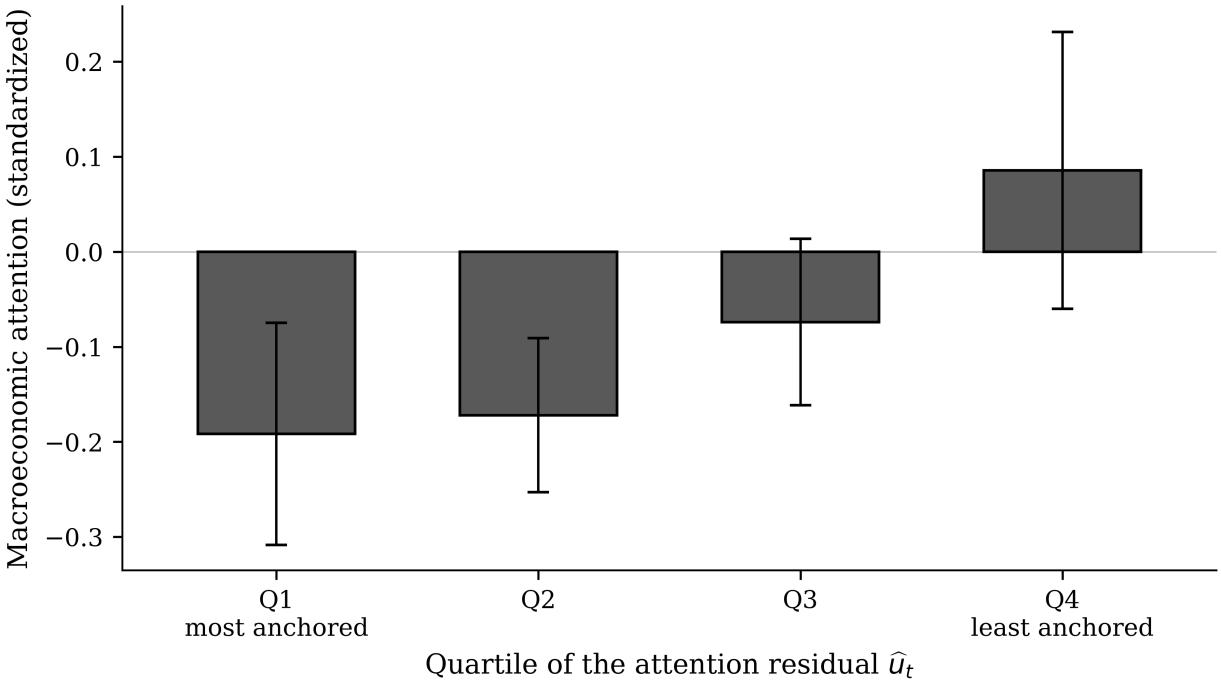


Figure 1: Macroeconomic attention across attention-state quartiles. The figure plots the mean of the standardized macroeconomic attention index of Fisher, Martineau, and Sheng (2022) within quartiles of the attention residual $\widehat{u}_t$, with 95% confidence intervals from Newey–West standard errors (five lags). Quartiles are formed from the five-day rolling median of $\widehat{u}$ relative to its trailing 252-day 25th, 50th, and 75th percentiles, classified at the prior day's close, so that Q1 collects the days on which institutional attention is most anchored on firm-specific news. Macroeconomic attention is lowest on the most anchored days and rises as attention detaches from firm-specific news, from −0.19 to 0.09 standard deviations across the quartiles. The sample is restricted to the 2,166 days on which the macroeconomic attention index is available.

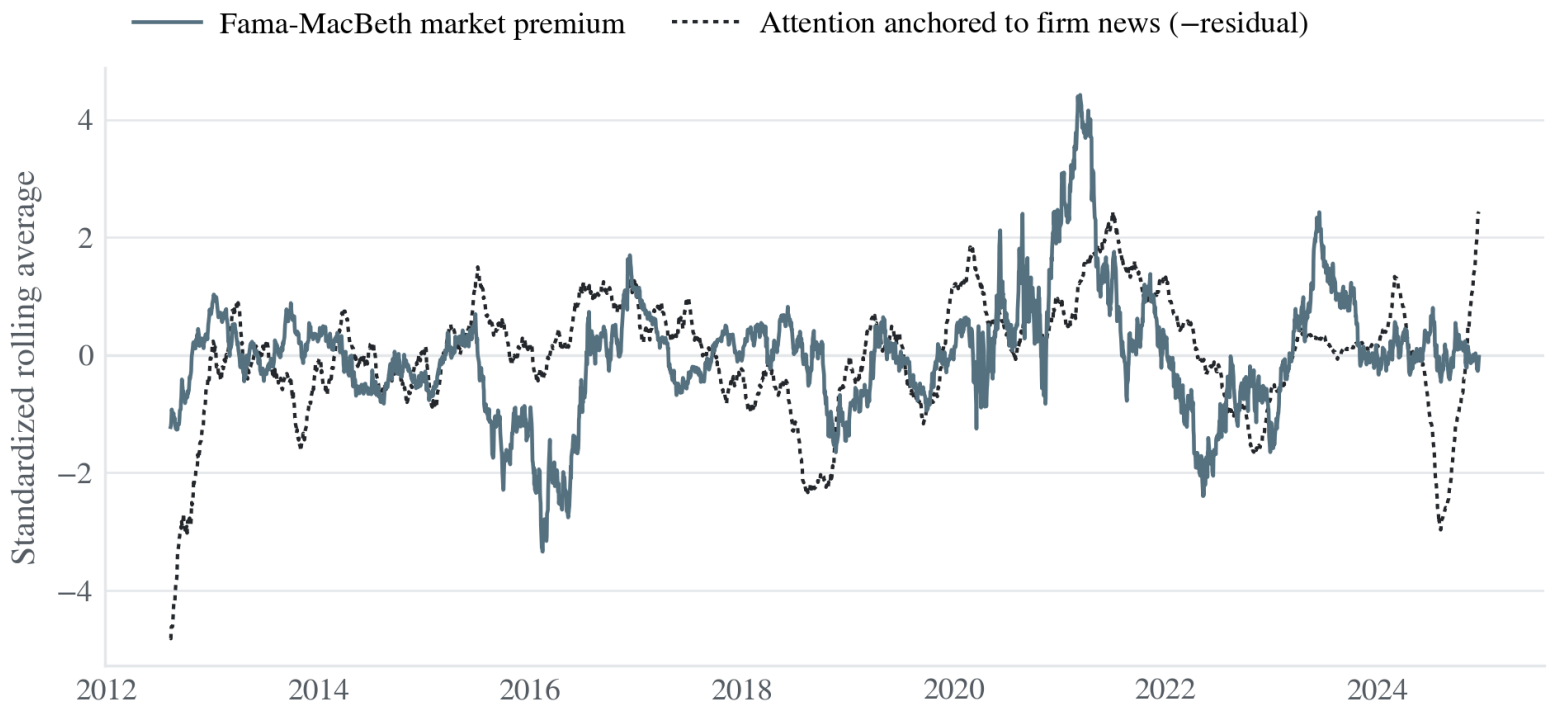


Figure 2: The market premium and attention composition over time. The solid line is the daily Fama-MacBeth slope on beta across the ten value-weighted beta-decile portfolios, averaged over a trailing 63-day window and standardized over the full sample. The dotted line is the attention composition residual $\hat{u}_t$, the expanding-window out-of-sample residual from regressing aggregate idiosyncratic attention on idiosyncratic news intensity, averaged and standardized in the same way. I plot $-\hat{u}_t$ so that higher values indicate attention anchored to firm-specific news, the state in which the model predicts beta is priced. The sample is May 2012 to December 2024. Both series are trailing averages, so turning points appear with a lag of roughly half the window; the later peak in $-\hat{u}_t$ in 2021 reflects the expanding regression absorbing the shift in attention composition during the retail trading episode rather than a lead of the premium over attention. At the daily frequency the cross-correlation between the two series is below 0.07 at every lead and lag from -20 to $+20$ days, and the premium does not predict next-day $\hat{u}_t$ once its own lag is controlled for.

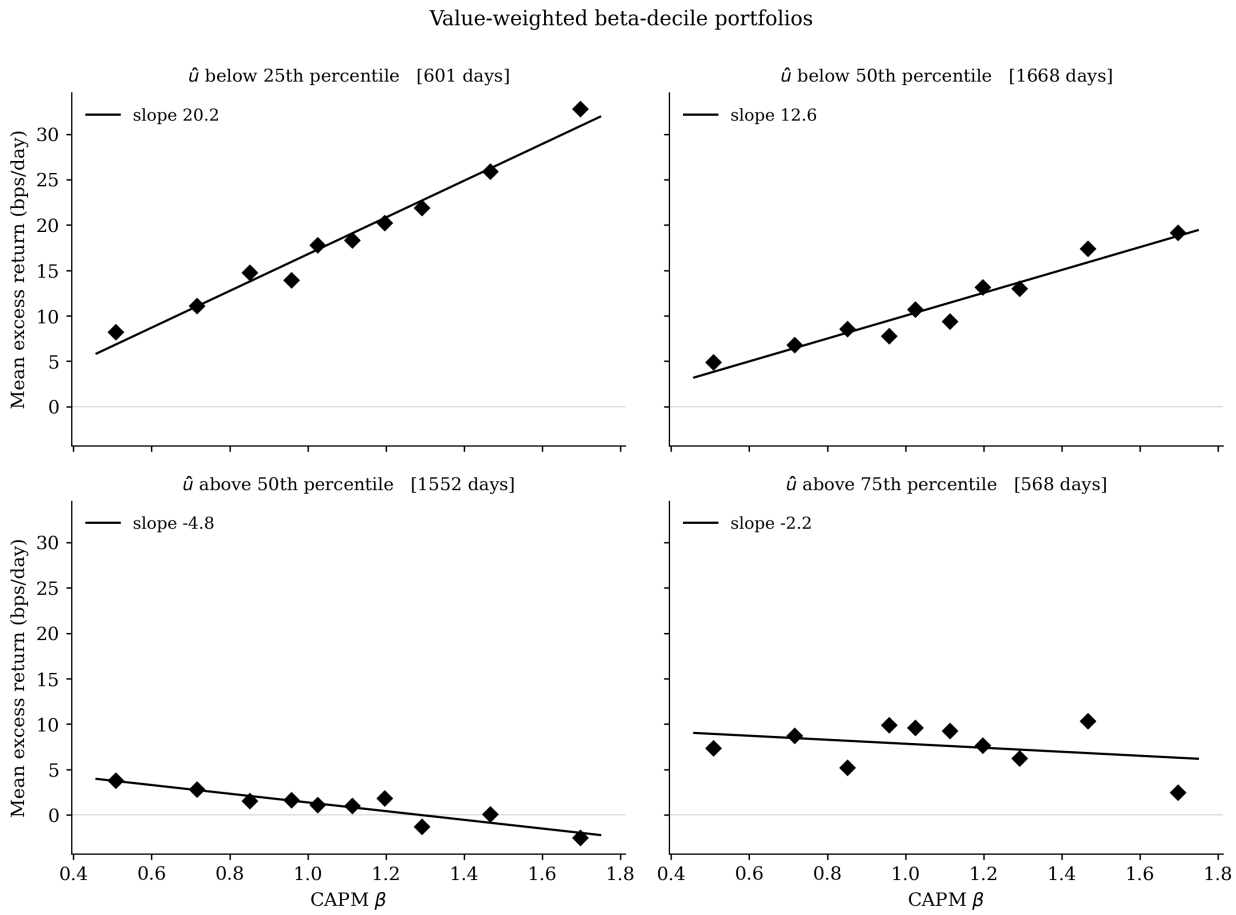


Figure 3: SMLs across residual attention states - **Value weighted portfolios**. This figure plots average portfolio excess returns against unconditional CAPM betas (Security Market Lines) for beta-sorted portfolios, separately by quantile cutoffs of the residual-based attention state. Panels correspond to state definitions based on rolling quantiles of the residual (0–25%, 0-50%, 50-100%, and 75-100%). Points show mean excess returns within each beta portfolio in the given state, and the solid line is the fitted linear relation between returns and betas within the state. The SML is steepest in low-residual states (0–25% and 0-50%), indicating a stronger positive risk–return tradeoff; it flattens or turns downward in high-residual states, consistent with weaker pricing of market risk when idiosyncratic attention explains less of market-wide attention.

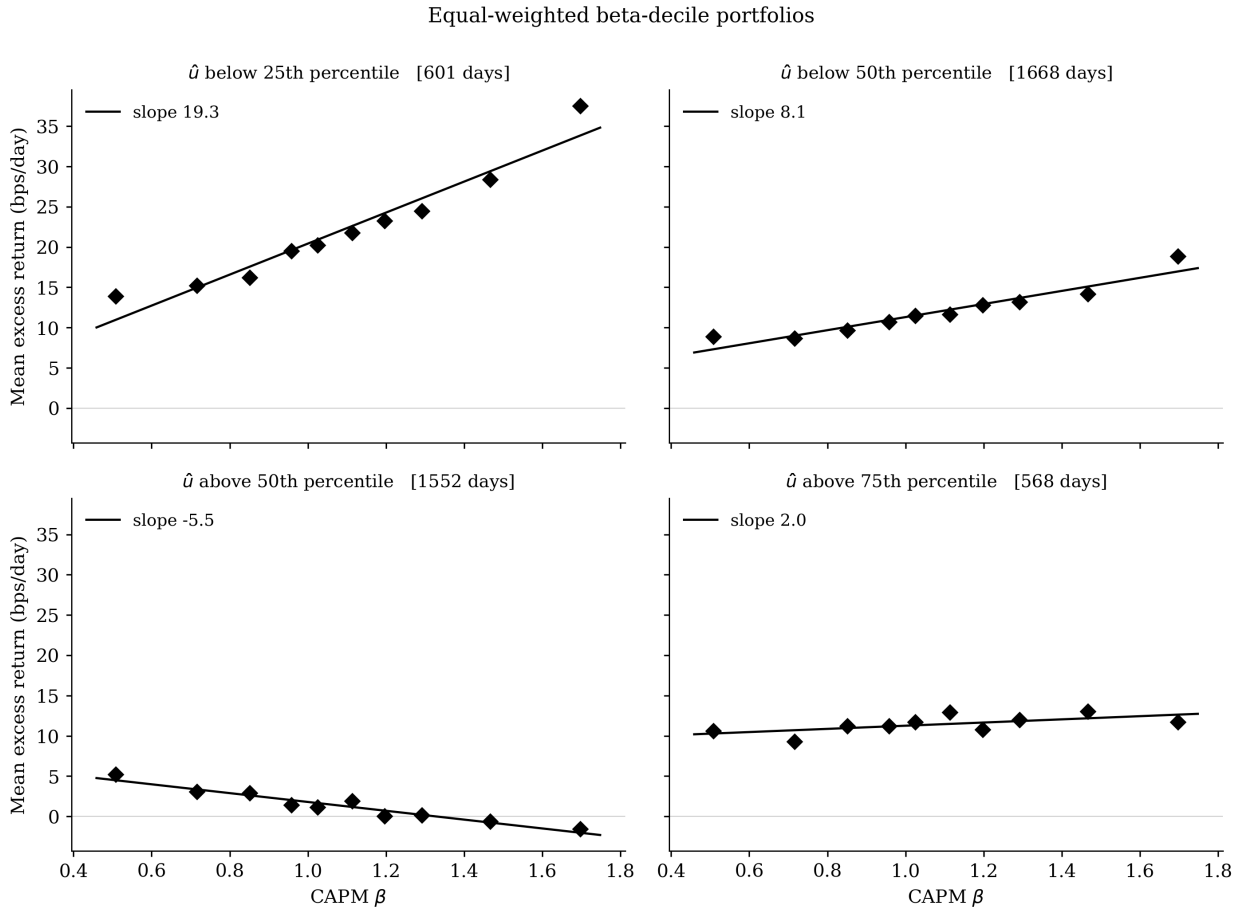


Figure 4: SMLs across residual attention states - **Equal weighted portfolios**. This figure plots average portfolio excess returns against unconditional CAPM betas (Security Market Lines) for beta-sorted portfolios, separately by quantile cutoffs of the residual-based attention state. Panels correspond to state definitions based on rolling quantiles of the residual (0–25%, 0-50%, 50-100%, and 75-100%). Points show mean excess returns within each beta portfolio in the given state, and the solid line is the fitted linear relation between returns and betas within the state. The SML is steepest in low-residual states (0–25% and 0-50%),indicating a stronger positive risk–return tradeoff; it flattens or turns downward in high-residual states, consistent with weaker pricing of market risk when idiosyncratic attention explains less of market-wide attention.

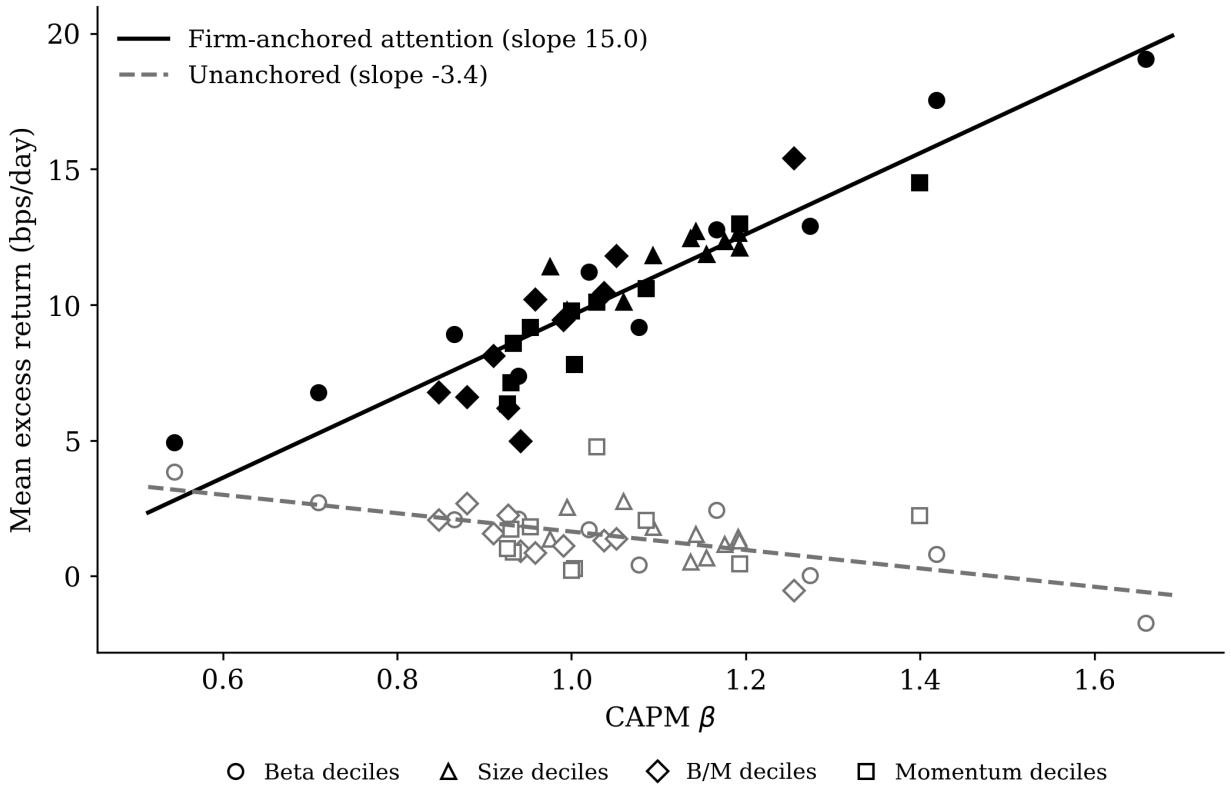


Figure 5: The security market line by attention state, all test-asset families. The figure plots mean daily excess returns (basis points per day) against CAPM betas for forty value-weighted portfolios: ten beta-sorted deciles (circles), ten size deciles (triangles), ten book-to-market deciles (diamonds), and ten momentum deciles (squares). Each portfolio's beta is estimated once over the full sample, so both states share the same horizontal positions and differ only in mean returns. Filled markers and the solid line correspond to days on which institutional attention is anchored on firm-specific news (the attention residual $\hat{u}$, smoothed over five days, lies below its trailing 252-day median, classified at the prior day's close); hollow markers and the dashed line correspond to the remaining days. Lines are ordinary least squares fits through the forty portfolio means within each state, with slopes reported in the legend. The sample covers May 2010 through January 2025.

A Proof of Lemma 1

The joint distribution of primitives and signals is multivariate Gaussian under Assumptions 1 and 2, so each posterior follows from conjugate normal updating.

Posterior of f. The only signal that depends on f is $z_j = f + \xi_j$. The prior density is proportional to $\exp(-\frac{1}{2}\tau_f f^2)$ and the likelihood to $\exp(-\frac{1}{2}\tau_\xi(z_j - f)^2)$, so the posterior is proportional to

$$\exp\Big(-\tfrac{1}{2}\big[\tau_f f^2 + \tau_\xi (z_j - f)^2\big]\Big).$$

Expanding and collecting terms in f, the exponent equals $(\tau_f + \tau_\xi) f^2 - 2\tau_\xi z_j f$ up to a constant, and completing the square gives

$$(\tau_f + \tau_\xi)\left[f - \frac{\tau_\xi z_j}{\tau_f + \tau_\xi}\right]^2 + \text{const.}$$

An exponent of the form $-\frac{1}{2}a(f-m)^2$ is the kernel of $\mathcal{N}(m, 1/a)$, hence $f \mid z_j \sim \mathcal{N}(\mu_{f,j}, V_f)$ with $V_f = (\tau_f + \tau_\xi)^{-1} = (\tau_f + (1-\omega)K)^{-1}$ and $\mu_{f,j} = V_f \tau_\xi z_j$. The public signals do not enter: under Assumption 1 they are independent of (f, z_j), so $p(f \mid z_j, y) = p(f \mid z_j)$.

Posterior of ε_i. The same computation with $y_i = \varepsilon_i + \eta_i$ in place of z_j gives $\varepsilon_i \mid y_i \sim \mathcal{N}(\mu_{\varepsilon,i}, V_\varepsilon)$ with $V_\varepsilon = (\tau_\varepsilon + \tau_y)^{-1} = (\tau_\varepsilon + \omega S)^{-1}$ and $\mu_{\varepsilon,i} = V_\varepsilon \tau_y y_i$.

Posterior of β_i. No signal depends on ν_i: y_i is independent of β_i by construction and z_j depends only on f. The posterior therefore equals the prior, $\beta_i \mid y_i \sim \mathcal{N}(b_i, \sigma_\nu^2)$.

Independence. Because $(f, \nu_i, \varepsilon_i)$ are mutually independent under the prior and each updates only through its own signal, the posteriors are mutually independent. Finally, $\mu_{f,j} = V_f \tau_\xi (f + \xi_j)$ is a linear function of Gaussian variables, and by the law of total variance $\mathrm{Var}(f) = \mathbb{E}[\mathrm{Var}(f \mid z_j)] + \mathrm{Var}(\mathbb{E}[f \mid z_j])$, so $\mathrm{Var}(\mu_{f,j}) = \sigma_f^2 - V_f$. □

B Posterior covariance and the residual variance $V_\varepsilon^\star$

From (11), write the forecast error of asset i as $a_i + b^{(i)} + c_i$ with $a_i = b_i(f - \mu_{f,j})$, $b^{(i)} = \nu_i f$, and $c_i = \varepsilon_i - \mu_{\varepsilon,i}$. The conditional covariance $\Sigma_{ik} = \mathbb{E}[(d_i - \mu_{j,i})(d_k - \mu_{j,k}) \mid \mathcal{I}_j]$ is the sum of nine cross-products:

$a_i a_k$:	$b_i b_k V_f$	since $\mathbb{E}[(f-\mu_{f,j})^2 \mid \mathcal{I}_j] = V_f$
$a_i b^{(k)}$:	0	$\mathbb{E}[(f-\mu_{f,j})\,\nu_k f \mid \mathcal{I}_j] = \mathbb{E}[\nu_k]\,\mathbb{E}[(f-\mu_{f,j}) f \mid \mathcal{I}_j] = 0$
$a_i c_k$:	0	f is independent of ε_k and η_k
$b^{(i)} b^{(k)}$:	$\sigma_\nu^2\,\mathbb{E}[f^2 \mid \mathcal{I}_j]$ if $i=k$, 0 otherwise	$\mathbb{E}[\nu_i \nu_k f^2 \mid \mathcal{I}_j] = \mathbb{E}[\nu_i \nu_k]\,\mathbb{E}[f^2 \mid \mathcal{I}_j]$
$b^{(i)} c_k$:	0	$\mathbb{E}[\nu_i] = 0$ and ν_i is independent of $(f, \varepsilon_k, \eta_k)$
$c_i c_k$:	V_ε if $i=k$, 0 otherwise	the $\varepsilon_i - \mu_{\varepsilon,i}$ are independent across assets

with $\mathbb{E}[f^2 \mid \mathcal{I}_j] = V_f + \mu_{f,j}^2$. Collecting terms gives (12).

The bracket in (12) depends on $\mu_{f,j}^2$ and hence on investor j's signal draw. Replacing $\mu_{f,j}^2$ by its ex-ante expectation and using $\mathrm{Var}(\mu_{f,j}) = \sigma_f^2 - V_f$ from Appendix A,

$$\mathbb{E}\Big[V_f + \mu_{f,j}^2\Big] = V_f + (\sigma_f^2 - V_f) = \sigma_f^2,$$

so the V_f terms cancel and the common diagonal term is $V_\varepsilon^\star = V_\varepsilon + \sigma_\nu^2 \sigma_f^2$, which is (13). In matrix form, $\Sigma = V_f\, bb' + V_\varepsilon^\star I$, whose diagonal entries $b_i^2 V_f + V_\varepsilon^\star$ measure the uncertainty remaining about asset i after learning and whose off-diagonal entries $b_i b_k V_f$ arise only because both assets load on the factor. □

C Proof of Lemma 2

From (9), investor j demands $q_j = \rho^{-1}\Sigma^{-1}(\mu_j - p)$. Since Σ and p are common across investors, aggregate demand is $\rho^{-1}\Sigma^{-1}(\bar{\mu} - p)$ with $\bar{\mu} = \int_0^1 \mu_j\, dj$. Integrating (10) term by term: $\int_0^1 \bar{d}\, dj = \bar{d}$; $\int_0^1 \mu_{\varepsilon,i}\, dj = \mu_{\varepsilon,i} = V_\varepsilon \tau_y y_i$, because every investor sees the same public news and processes it with the same precision; and $\int_0^1 b_i \mu_{f,j}\, dj = b_i V_f \tau_\xi \int_0^1 (f + \xi_j)\, dj = b_i V_f \tau_\xi f$, because the private noise is independent across investors with mean zero and averages to zero over the continuum. Hence $\bar{\mu}_i = \bar{d} + b_i \bar{\mu}_f + V_\varepsilon \tau_y y_i$ with $\bar{\mu}_f = V_f \tau_\xi f$.

Market clearing sets aggregate demand equal to supply, $\rho^{-1}\Sigma^{-1}(\bar{\mu} - p) = \bar{x}\mathbf{1} + u$, so $p = \bar{\mu} - \rho\Sigma(\bar{x}\mathbf{1} + u)$. The ith entry of Σv is $\sum_k \Sigma_{ik} v_k$. Using (14),

$$[\Sigma\, \bar{x}\mathbf{1}]_i = \sum_k \Big(b_i b_k V_f + V_\varepsilon^\star \mathbf{1}\{i=k\}\Big)\, \bar{x} = V_f\, b_i\, (N\bar{b}\, \bar{x}) + V_\varepsilon^\star\, \bar{x},$$
$$[\Sigma u]_i = \sum_k \Big(b_i b_k V_f + V_\varepsilon^\star \mathbf{1}\{i=k\}\Big)\, u_k = V_f\, b_i\, (b'u) + V_\varepsilon^\star\, u_i,$$

using $\sum_k b_k = N\bar{b}$ and $b'u \equiv \sum_k b_k u_k$. Substituting into $p_i = \bar{\mu}_i - \rho[\Sigma(\bar{x}\mathbf{1} + u)]_i$ gives (17). □

D Derivation of the Fama–MacBeth slope and attenuation factor

From (18), the expected excess return of asset i is $Y_i \equiv \mathbb{E}[d_i - p_i] = \lambda^A b_i + \kappa$ with $\lambda^A \equiv \rho V_f N \bar{b} \bar{x}$ and $\kappa \equiv \rho V_\varepsilon^\star \bar{x}$. A cross-sectional regression of Y_i on the true prior loading b_i would identify λ^A.

The econometrician instead regresses on $\hat{\beta}_i = b_i + \nu_i + e_i$, where e_i is the sampling error of a T-period rolling OLS regression of realized returns on the factor. Realized return variance has a factor component $\beta_i^2 \sigma_f^2$ and a non-factor component $\sigma_\varepsilon^2 + 1/\tau_y$, the latter combining the idiosyncratic shock and the public-signal noise that remains in the return, so standard OLS algebra gives $\mathrm{Var}(e_i) \approx (\sigma_\varepsilon^2 + 1/\tau_y)/(T\sigma_f^2) = V_e(\omega)$, which is decreasing in ω because $\tau_y = \omega S$.

Treating b_i as a cross-sectional draw with variance σ_b^2, independent of ν_i and e_i, the constant κ drops out of the covariance and

$$\mathrm{Cov}_i(Y_i, \hat{\beta}_i) = \lambda^A \Big(\mathrm{Cov}(b_i, b_i) + \mathrm{Cov}(b_i, \nu_i) + \mathrm{Cov}(b_i, e_i) \Big) = \lambda^A \sigma_b^2, \qquad \mathrm{Var}_i(\hat{\beta}_i) = \sigma_b^2 + \sigma_\nu^2 + V_e = s_\beta^2 + V_e.$$

Therefore

$$\lambda^{FM} = \frac{\mathrm{Cov}_i(Y_i, \hat{\beta}_i)}{\mathrm{Var}_i(\hat{\beta}_i)} = \lambda^A \cdot \frac{\sigma_b^2}{s_\beta^2 + V_e} = \lambda^A \cdot \frac{s_\beta^2 - \sigma_\nu^2}{s_\beta^2 + V_e} = A(\omega)\, \lambda^A,$$

which is (19). Differentiating $A(\omega)$ with respect to ω holding s_β^2 and σ_ν^2 fixed gives $dA/d\omega = -\sigma_b^2 \, (s_\beta^2 + V_e)^{-2} \, dV_e/d\omega > 0$, the measurement channel in (21).

With realized rather than expected returns as the regressand, $d_i - p_i = Y_i + w_i$ where w_i is a mean-zero surprise, so on a single day the estimated slope equals $A\lambda^A$ plus a term driven by that day's realizations of the factor and of supply. Averaging the daily slopes removes this term provided the attention state does not forecast the market and the beta window ends before the return day, so the Fama–MacBeth average of realized-return slopes is unbiased for λ^{FM}. □

E Optimal attention

The main text takes the attention share ω as given, because Propositions 1 and 2 hold for any ω. This appendix endogenizes it and shows that the investor's problem has an interior solution, so that the equilibrium studied in the text is one in which some attention is devoted to each channel.

Value of information. Under CARA-Gaussian preferences, the ex-ante certainty-equivalent gain from an information set is proportional to $\frac{1}{2}\log\det\left(\Sigma_0\,\Sigma(\omega)^{-1}\right)$, where Σ_0 is the prior covariance of payoffs and $\Sigma(\omega)$ the posterior covariance under attention share ω. The prior term does not depend on ω. By the rank-one-plus-diagonal structure in (14) and the matrix determinant lemma,

$$\det\Sigma(\omega) = \left(V_\varepsilon^\star(\omega)\right)^{N-1}\left(V_\varepsilon^\star(\omega) + V_f(\omega)\textstyle\sum_i b_i^2\right),$$

so, dropping constants, the investor chooses $\omega \in [0,1]$ to maximize

$$\mathcal{V}(\omega) = -\frac{N-1}{2}\,\log V_\varepsilon^\star(\omega) - \frac{1}{2}\,\log\!\left(V_\varepsilon^\star(\omega) + V_f(\omega)\textstyle\sum_i b_i^2\right). \tag{27}$$

The first term is the gain from reducing residual uncertainty across the N assets; the second is the cost of bearing factor risk, which rises with the factor eigenvalue of the posterior covariance.

Derivatives. From (13), $V_\varepsilon^\star = V_\varepsilon + \sigma_\nu^2\sigma_f^2$ with the second term constant in ω, so $dV_\varepsilon^\star/d\omega = dV_\varepsilon/d\omega = -SV_\varepsilon^2 < 0$: attention to firm news lowers residual uncertainty. From Proposition 1, $dV_f/d\omega = KV_f^2 > 0$. Hence

$$\mathcal{V}'(\omega) = \frac{(N-1)\,SV_\varepsilon^2}{2V_\varepsilon^\star} - \frac{1}{2}\,\frac{-SV_\varepsilon^2 + KV_f^2\textstyle\sum_i b_i^2}{V_\varepsilon^\star + V_f\textstyle\sum_i b_i^2}. \tag{28}$$

The first term is the marginal benefit of firm-news attention, positive and proportional to the number of assets. The second collects the marginal cost, in which $KV_f^2\sum_i b_i^2$ is the increase in factor risk from learning less about f, partly offset by the reduction in residual risk.

Proposition 3 (Interior optimum)**.** *For all admissible parameter values $(N, S, K, \rho, \sigma_f, \sigma_\nu, \sigma_\varepsilon, \tau_f)$ with $N \geq 2$, $S > 0$, $K > 0$, and $\sum_i b_i^2 > 0$, there exists an attention share $\omega^\star \in (0,1)$ satisfying $\mathcal{V}'(\omega^\star) = 0$.*

Proof. $\mathcal{V}$ is continuous on $[0,1]$ and differentiable on $(0,1)$. As $\omega \to 0^+$, $V_\varepsilon \to \sigma_\varepsilon^2$ and $V_f \to (\tau_f + K)^{-1}$, so the marginal benefit in (28) is $(N-1)S\sigma_\varepsilon^4/(2V_\varepsilon^\star) > 0$ while the cost term is bounded; for N large relative to $KV_f^2\sum_i b_i^2/V_\varepsilon^\star$, or more generally whenever the benefit of the first unit of firm-news attention exceeds its factor-risk cost, $\mathcal{V}'(0^+) > 0$. As $\omega \to 1^-$, $V_\varepsilon \to (\tau_\varepsilon + S)^{-1}$, so the benefit term SV_ε^2 is small, while $V_f \to \sigma_f^2$ so the cost term $K\sigma_f^4\sum_i b_i^2$ is at its largest; for any $K > 0$ and non-degenerate cross-section, $\mathcal{V}'(1^-) < 0$. By the intermediate value theorem, $\mathcal{V}'(\omega^\star) = 0$ for some $\omega^\star \in (0,1)$. □

The interior optimum reflects the trade-off at the heart of the model. Moving attention toward firm-specific news improves idiosyncratic learning, but because firm news is orthogonal to the factor under Assumption 1, every unit reallocated to public news is a unit lost from factor learning, and the investor balances the two. Across days, the optimum moves with the primitives the model takes as given: a larger supply of firm news S raises the marginal benefit of attending to it, and greater private research capacity K raises the cost of neglecting the factor. The model does not pin down the daily determinants of $\omega^\star$, which is why the empirical work measures the realized allocation directly rather than imputing it from primitives.